



Unit - 1 - lifestyle disease notes

Biomedical engineering (Anna University)



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OBT356 - LIFESTYLE DISEASES

UNIT I INTRODUCTION

9

Lifestyle diseases – Definition; Risk factors – Eating, smoking, drinking, stress, physical activity, illicit drug use; Obesity, diabetes, cardiovascular diseases, respiratory diseases, cancer; Prevention – Diet and exercise.

UNIT II CANCER

9

Types - Lung cancer, Mouth cancer, Skin cancer, cervical cancer, Carcinoma oesophagus; Causes Tobacco usage, Diagnosis – Biomarkers, Treatment

UNIT III CARDIOVASCULAR DISEASES

9

Coronary atherosclerosis – Coronary artery disease; Causes -Fat and lipids, Alcohol abuse -- Diagnosis - Electrocardiograph, echocardiograph, Treatment, Exercise and Cardiac rehabilitation

UNIT IV DIABETES AND OBESITY

9

Types of Diabetes mellitus; Blood glucose regulation; Complications of diabetes – Pediatric and Adolescent obesity – Weight control and BMI

UNIT V RESPIRATORY DISEASES

9

Chronic lung disease, Asthma, COPD; Causes - Breathing pattern (Nasal vs mouth), Smoking – Diagnosis - Pulmonary function testing

TOTAL: 45 PERIODS

TEXT BOOKS:

1. R.Kumar & Meenal Kumar, “Guide to Prevention of Lifestyle Diseases”, Deep & Deep Publications, 2003
2. Gary Eggar et al, “Lifestyle Medicine”, 3rd Edition, Academic Press, 2017

REFERENCES:

1. James M.R, “Lifestyle Medicine”, 2nd Edition, CRC Press, 2013
2. Akira Miyazaki et al, “New Frontiers in Lifestyle-Related Disease”, Springer, 2008

UNIT – I INTRODUCTION

1. INTRODUCTION

Lifestyle diseases – Definition; Risk factors – Eating, smoking, drinking, stress, physical activity, illicit drug use; Obesity, diabetes, cardiovascular diseases, respiratory diseases, cancer; Prevention – Diet and exercise.

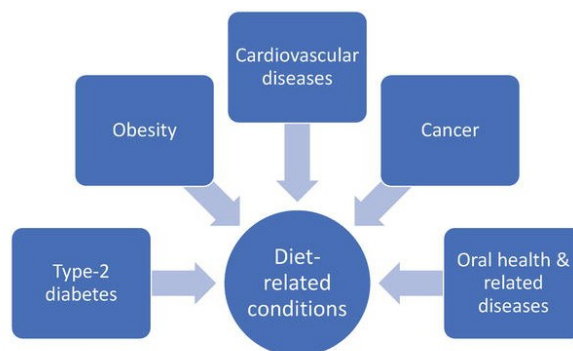
1.1 Life style diseases

Objective: The students will learn the various diseases and its effects.

Definition

Lifestyle diseases, also known as non-communicable diseases (NCDs), refer to diseases that are primarily caused by unhealthy lifestyle choices. These diseases are chronic in nature and generally develop over an extended period.

They are often linked to factors such as poor diet, lack of physical activity, excessive alcohol consumption, and tobacco use. Examples of lifestyle diseases include type 2 diabetes, cardiovascular diseases (like heart disease and stroke), obesity, certain cancers (such as lung cancer from smoking), and chronic respiratory diseases (like chronic obstructive pulmonary disease from smoking). The term "lifestyle diseases" emphasizes the role of individual behavior and choices in their prevention and management.



1.2 Risk factors

Objective: The students will learn the various Risk factors and its effects of diseases.

- Cigarette and tobacco smoke
- High blood cholesterol
- High blood pressure
- Physical inactivity
- Obesity and overweight
- Diabetes mellitus

This emphasis on prevention has the potential to produce better health as well as to offer tremendous cost savings. We are designed to be predominantly vegetarian creatures. Our present culture creates a bounty of meat, dairy products, sweets and fats. As a direct consequence of the above we get increasing skin fold thicknesses, hypertension, coronary artery disease, cancers and many other disabling and killing ailments.



A few Basic Errors are given below:

- (i) The consumption of smoke and pollutants: The human body needs to breathe clean air and not smoke chemicals or polluted or stale air.
- (ii) The consumption of alcohol, colas and caffeine: The human body needs clean and sterile water.
- (iii) The dietary consumption of an abundance of fats, animal proteins and sweets: The human body needs high fibre, whole grains, vegetables and beans and fruits with small amounts of fish, nuts and seeds.
- (iv) The neglect of exercise: The human body is designed to be lean and muscular, with active life and avoiding sedentary life and too much comfort. Diseases such as diverticulitis, kidney failure, dialysis, gallstones, diabetes, osteoporosis, hypertension, coronary artery disease, stroke, and numerous cancers, including breast, uterine, prostate and colon cancer, are and can be publicly viewed as virtually preventable with exercise and other measures.

1.2.1 Eating

Lifestyle diseases, often referred to as non-communicable diseases (NCDs), are ailments that primarily result from lifestyle choices and habits. They include heart disease, stroke, obesity, type 2 diabetes, and certain types of cancer. Diet plays a significant role in either mitigating or exacerbating these conditions. Here are some dietary factors related to lifestyle diseases:

1.2.1.1 Common Dietary Factors Contributing to Lifestyle Diseases

1. **High Sugar Intake:**

- **Impact:** Excessive consumption of sugar, especially from sugary drinks and snacks, can lead to obesity and type 2 diabetes.
- **Sources:** Soda, candy, baked goods, and many processed foods.

2. **High Fat Intake:**

- **Impact:** Diets high in unhealthy fats (trans fats and saturated fats) can increase the risk of heart disease and stroke.
- **Sources:** Fried foods, processed snacks, and fatty cuts of meat.

3. **High Salt Intake:**

- **Impact:** Excessive salt consumption can lead to high blood pressure, which is a risk factor for heart disease and stroke.
- **Sources:** Processed foods, canned soups, and restaurant meals.

4. **Low Fiber Intake:**

- **Impact:** Diets low in fiber are linked to digestive issues, type 2 diabetes, and heart disease.
- **Sources:** Whole grains, fruits, vegetables, and legumes.

5. **Low Fruit and Vegetable Intake:**

- **Impact:** Insufficient consumption of fruits and vegetables can lead to a deficiency in essential vitamins and minerals, contributing to various lifestyle diseases.
- **Sources:** A variety of fresh, frozen, or canned fruits and vegetables.

1.2.1.2 Dietary Recommendations to Prevent Lifestyle Diseases

1. **Increase Intake of Whole Foods:**

- **Action:** Focus on whole grains, fresh fruits, and vegetables.
- **Benefits:** These foods are high in essential nutrients and fiber, which can help prevent obesity, heart disease, and diabetes.

2. **Reduce Processed Food Consumption:**

- **Action:** Limit foods that are high in sugar, unhealthy fats, and salt.
- **Benefits:** Helps maintain healthy weight and reduce risks of heart disease and hypertension.

3. **Choose Healthy Fats:**

- **Action:** Opt for fats from sources like avocados, nuts, seeds, and fish.
- **Benefits:** These fats can help reduce the risk of heart disease.

4. **Moderate Portion Sizes:**

- **Action:** Be mindful of portion sizes to avoid overeating.
- **Benefits:** Helps in maintaining a healthy weight and preventing obesity-related diseases.

5. **Stay Hydrated:**

- **Action:** Drink plenty of water and avoid sugary drinks.
- **Benefits:** Proper hydration supports overall health and helps prevent overeating.

6. **Limit Red and Processed Meats:**

- **Action:** Choose lean protein sources like poultry, fish, beans, and legumes.
- **Benefits:** Reduces the risk of certain cancers and heart disease.

By making informed dietary choices and maintaining a balanced diet, individuals can significantly reduce their risk of developing lifestyle diseases.

1.2.2 Smoking

Smoking is a major contributor to various lifestyle diseases and has significant negative impacts on health. Here are some key points about smoking and its effects:

1.2.2.1 Health Impacts of Smoking

1. **Cardiovascular Diseases:**

- **Impact:** Smoking damages blood vessels, increases blood pressure, and lowers oxygen in the blood, leading to an increased risk of heart disease, stroke, and peripheral artery disease.
- **Mechanism:** Nicotine and other chemicals in cigarettes cause the arteries to narrow and harden, making it harder for blood to flow.

2. **Respiratory Diseases:**

- **Impact:** Smoking is a primary cause of chronic obstructive pulmonary disease (COPD), which includes emphysema and chronic bronchitis.
- **Mechanism:** The chemicals in tobacco smoke damage the airways and air sacs in the lungs.

3. **Cancer:**

- **Impact:** Smoking is a leading cause of various cancers, including lung, throat, mouth, esophagus, pancreas, bladder, kidney, and cervix.
- **Mechanism:** Carcinogens in tobacco smoke cause mutations in DNA, leading to uncontrolled cell growth.

4. **Diabetes:**

- **Impact:** Smokers are at a higher risk of developing type 2 diabetes.
- **Mechanism:** Smoking causes inflammation and oxidative stress, impairing insulin function.

5. **Reproductive Health:**

- **Impact:** Smoking can cause infertility, complications during pregnancy, and can harm the health of the fetus.
- **Mechanism:** Nicotine and other chemicals affect hormone production and blood flow to reproductive organs.

6. **Weakened Immune System:**

- **Impact:** Smoking weakens the immune system, making the body more susceptible to infections and diseases.
- **Mechanism:** The chemicals in tobacco smoke interfere with the immune system's ability to fight off infections.

1.2.2.2 **Benefits of Quitting Smoking**

1. **Improved Heart Health:**

- **Benefit:** Reduces the risk of heart disease and stroke.
- **Timeline:** Heart rate and blood pressure drop within 20 minutes of quitting, and the risk of heart disease is cut in half after one year.

2. **Better Lung Function:**

- **Benefit:** Slows the progression of lung damage and reduces the risk of lung diseases.
- **Timeline:** Lung function improves within a few months, and the risk of lung cancer is significantly reduced after 10 years.

3. **Lower Cancer Risk:**

- **Benefit:** Reduces the risk of various cancers over time.
- **Timeline:** The risk of mouth, throat, esophagus, and bladder cancers is cut in half within five years.

4. **Enhanced Reproductive Health:**

- **Benefit:** Improves fertility and pregnancy outcomes.
- **Timeline:** Positive effects can be seen almost immediately after quitting.

5. **Stronger Immune System:**

- **Benefit:** Better resistance to infections and faster recovery from illnesses.
- **Timeline:** Immune function begins to improve within weeks of quitting.

6. **Overall Well-being:**

- **Benefit:** Better sense of taste and smell, more energy, and improved physical fitness.
- **Timeline:** Benefits are noticeable within days to weeks after quitting.

1.2.2.3 Strategies for Quitting Smoking

1. Nicotine Replacement Therapy (NRT):

- **Options:** Patches, gum, lozenges, inhalers, and nasal sprays.
- **Benefit:** Helps reduce withdrawal symptoms by providing a lower dose of nicotine without the harmful chemicals in tobacco smoke.

2. Medications:

- **Options:** Prescription medications like bupropion (Zyban) and varenicline (Chantix).
- **Benefit:** Helps reduce cravings and withdrawal symptoms.

3. Behavioral Therapy:

- **Options:** Counseling, support groups, and cognitive-behavioral therapy.
- **Benefit:** Provides strategies to cope with triggers and develop healthier habits.

4. Lifestyle Changes:

- **Options:** Regular physical activity, healthy diet, and stress management techniques.
- **Benefit:** Helps manage cravings and improve overall health.

5. Support Systems:

- **Options:** Family, friends, and smoking cessation programs.
- **Benefit:** Emotional support and encouragement increase the chances of quitting successfully.

1.2.3 Alcohol drinking

Alcohol drinking refers to the consumption of beverages containing ethanol, commonly known as alcohol. This practice is widespread across many cultures and is often associated with social events, celebrations, and relaxation. However, drinking alcohol can have both short-term and long-term effects on health and behavior.

Short-Term Effects

- **Euphoria and relaxation:** Small amounts of alcohol can create feelings of happiness and relaxation.
- **Impaired judgment and coordination:** As blood alcohol levels rise, judgment and motor skills can be affected, increasing the risk of accidents and injuries.
- **Slurred speech and drowsiness:** Higher levels of consumption can lead to noticeable impairment, including difficulty speaking and extreme tiredness.
- **Alcohol poisoning:** Consuming large quantities in a short period can be life-threatening, leading to symptoms like confusion, vomiting, seizures, slow breathing, and unconsciousness.

Long-Term Effects

- **Addiction:** Chronic alcohol use can lead to dependence and addiction, known as alcoholism.
- **Liver disease:** Prolonged heavy drinking can cause liver damage, including fatty liver, hepatitis, and cirrhosis.
- **Cardiovascular problems:** Excessive alcohol consumption can increase the risk of high blood pressure, heart disease, and stroke.
- **Mental health issues:** Long-term alcohol use is linked to depression, anxiety, and other mental health disorders.
- **Increased cancer risk:** Alcohol consumption is associated with a higher risk of several types of cancer, including mouth, throat, esophagus, liver, colon, and breast cancer.

Safe Drinking Guidelines

- **Moderation:** Many health organizations recommend drinking in moderation. For example, in the United States, moderate drinking is defined as up to one drink per day for women and up to two drinks per day for men.
- **Understanding standard drinks:** A standard drink contains about 14 grams of pure alcohol, which is roughly equivalent to:
 - 12 ounces of beer (5% alcohol content)
 - 5 ounces of wine (12% alcohol content)
 - 1.5 ounces of distilled spirits (40% alcohol content)

- **Avoiding binge drinking:** Binge drinking is defined as consuming a large amount of alcohol in a short period, typically five or more drinks for men and four or more drinks for women within about two hours.

Risks and Considerations

- **Pregnancy:** Drinking alcohol during pregnancy can lead to fetal alcohol spectrum disorders (FASD), causing physical, behavioral, and learning problems in the child.
- **Driving:** Alcohol impairs driving ability, and it's illegal and dangerous to drive under the influence.
- **Medication interactions:** Alcohol can interact negatively with various medications, reducing their effectiveness or causing harmful side effects.

1.2.4 Stress

Stress is a natural response to challenging or demanding situations, and it can have both positive and negative effects on individuals. While a certain level of stress can motivate and enhance performance, excessive or chronic stress can lead to various physical and mental health issues.

1.2.4.1 Causes of Stress

Stress can arise from various sources, commonly referred to as stressors, which can be categorized into different types:

- **Work-related stress:** Deadlines, high workloads, job insecurity, and conflicts with colleagues or supervisors.
- **Financial stress:** Debt, lack of savings, and concerns about financial stability.
- **Relationship stress:** Conflicts with family, friends, or romantic partners.
- **Life changes:** Major events such as moving, getting married, or experiencing a loss.
- **Health-related stress:** Illness, injury, or concerns about one's health or the health of loved ones.
- **Environmental stress:** Noise, pollution, or living in an unsafe neighborhood.

Physical and Emotional Symptoms of Stress

Stress can manifest in various ways, affecting both the body and the mind:

Physical symptoms:

- Headaches
- Muscle tension or pain

- Fatigue
- Sleep disturbances
- Digestive issues
- Changes in appetite

Emotional and psychological symptoms:

- Anxiety
- Irritability or anger
- Sadness or depression
- Restlessness
- Difficulty concentrating
- Feeling overwhelmed

Coping Strategies

Effective stress management involves a combination of techniques and lifestyle changes to reduce stress and improve overall well-being:

- **Exercise:** Regular physical activity can reduce stress hormones and promote relaxation.
- **Healthy diet:** Eating a balanced diet can improve energy levels and overall health.
- **Sleep:** Prioritizing good sleep hygiene can help the body and mind recover from stress.
- **Mindfulness and relaxation techniques:** Practices such as meditation, deep breathing, and yoga can calm the mind and reduce stress.
- **Time management:** Organizing and prioritizing tasks can reduce the feeling of being overwhelmed.
- **Social support:** Talking to friends, family, or a mental health professional can provide emotional support and different perspectives on stressors.
- **Hobbies and leisure activities:** Engaging in enjoyable activities can provide a mental break from stress.
- **Avoiding unhealthy coping mechanisms:** Reducing the use of alcohol, drugs, or other unhealthy habits can prevent further stress and health issues.

1.2.5 Physical activities

Physical activities are any bodily movements produced by skeletal muscles that require energy expenditure. They range from simple tasks like walking and household chores

to more structured exercises and sports. Here are some categories and examples of physical activities:

Aerobic Activities

- **Walking:** A low-impact exercise suitable for all fitness levels.
- **Running/Jogging:** Higher intensity than walking, beneficial for cardiovascular health.
- **Cycling:** Can be done indoors on a stationary bike or outdoors.
- **Swimming:** Full-body workout that's easy on the joints.
- **Dancing:** Fun and social way to get moving.

Strength Training

- **Weightlifting:** Using free weights or weight machines to build muscle.
- **Bodyweight Exercises:** Push-ups, pull-ups, squats, and lunges.
- **Resistance Bands:** Portable and versatile for various exercises.

Flexibility and Balance

- **Yoga:** Combines physical postures, breathing exercises, and meditation.
- **Pilates:** Focuses on core strength, flexibility, and overall body awareness.
- **Tai Chi:** Gentle martial arts form that enhances balance and relaxation.

Sports and Recreational Activities

- **Team Sports:** Soccer, basketball, volleyball, and others that involve group play.
- **Racquet Sports:** Tennis, badminton, squash, and ping pong.
- **Outdoor Activities:** Hiking, rock climbing, kayaking, and skiing.
- **Martial Arts:** Karate, judo, taekwondo, and other self-defense disciplines.

Daily Activities

- **Household Chores:** Cleaning, gardening, and other home maintenance tasks.
- **Commuting:** Walking or cycling to work or school.
- **Playing with Kids or Pets:** Active playtime that involves running or chasing.

Benefits of Physical Activity

- **Improves Cardiovascular Health:** Reduces the risk of heart disease, stroke, and high blood pressure.
- **Enhances Muscle and Bone Strength:** Helps prevent osteoporosis and muscle loss.
- **Aids in Weight Management:** Burns calories and boosts metabolism.
- **Boosts Mental Health:** Reduces symptoms of depression and anxiety, improves mood and cognitive function.

- **Increases Longevity:** Associated with a longer, healthier life.

Tips for Incorporating Physical Activity

- **Set Realistic Goals:** Start with small, achievable targets and gradually increase intensity.
- **Find Activities You Enjoy:** This makes it easier to stick with a regular routine.
- **Stay Consistent:** Aim for at least 150 minutes of moderate aerobic activity or 75 minutes of vigorous activity per week, along with muscle-strengthening exercises twice a week.
- **Mix It Up:** Variety prevents boredom and works different muscle groups.
- **Listen to Your Body:** Rest when needed and avoid overtraining to prevent injuries.

General Guidelines for Incorporating Physical Activity

- **Consult a Healthcare Provider:** Especially important if you have a chronic condition or are new to exercise.
- **Start Slowly:** Gradually increase the intensity and duration of your workouts.
- **Set Realistic Goals:** Establish achievable targets to stay motivated.
- **Mix It Up:** Incorporate a variety of exercises to keep your routine interesting and work different muscle groups.
- **Stay Consistent:** Aim for at least 150 minutes of moderate-intensity aerobic activity or 75 minutes of vigorous-intensity activity per week, along with muscle-strengthening activities on two or more days a week.
- **Monitor Your Progress:** Keep track of your activities and how you feel to stay motivated and adjust your routine as needed.

By integrating regular physical activity into your lifestyle, you can significantly reduce the risk of lifestyle diseases and improve overall health and well-being. Regular physical activity is crucial for maintaining overall health and well-being. Whether you prefer structured workouts or integrating more movement into your daily routine, finding ways to stay active can lead to a happier and healthier life.

1.2.6 Illicit Drug Use

Illicit drug use refers to the consumption of substances that are illegal to possess, distribute, or use. These substances can have profound negative effects on physical and

mental health, social relationships, and overall well-being. Commonly misused illicit drugs include cocaine, heroin, methamphetamine, ecstasy (MDMA), LSD, and marijuana (in places where it remains illegal).

Health Risks of Illicit Drug Use

1. Physical Health

- **Cardiovascular Issues:** Drugs like cocaine and methamphetamine can cause heart attacks, strokes, and high blood pressure.
- **Respiratory Problems:** Smoking drugs can lead to lung damage and respiratory infections.
- **Liver Damage:** Hepatitis and other liver conditions are common among users of injected drugs.
- **Infectious Diseases:** Sharing needles can lead to HIV, hepatitis B and C, and other infections.
- **Overdose:** Many illicit drugs carry a high risk of overdose, which can be fatal.

2. Mental Health

- **Addiction:** Illicit drugs are often highly addictive, leading to physical and psychological dependence.
- **Mental Disorders:** Substance abuse can exacerbate or trigger mental health conditions like depression, anxiety, and psychosis.
- **Cognitive Impairment:** Long-term use can impair memory, attention, and decision-making skills.

3. Social and Legal Consequences

- **Relationships:** Drug use can strain or destroy relationships with family, friends, and colleagues.
- **Employment:** Job loss or difficulty maintaining employment due to drug use and related behaviors.
- **Legal Issues:** Arrests, fines, and imprisonment for possession, distribution, or use of illegal substances.
- **Financial Problems:** The cost of drugs and legal issues can lead to significant financial instability.

Prevention and Treatment

Prevention

1. **Education:** Increasing awareness about the dangers of drug use through school programs, community outreach, and public health campaigns.

2. **Community Support: Strengthening community ties and providing resources for at-risk populations.**
3. **Early Intervention: Identifying and supporting individuals at risk of substance abuse before it starts.**

Treatment

1. **Detoxification:** Medical supervision to safely manage withdrawal symptoms.
2. **Rehabilitation Programs:** Inpatient or outpatient programs that provide comprehensive treatment, including therapy and medical care.
3. **Therapy and Counseling:** Cognitive-behavioral therapy (CBT), motivational interviewing, and other forms of counseling to address underlying issues and promote recovery.
4. **Medication-Assisted Treatment (MAT):** Using medications to reduce cravings and withdrawal symptoms (e.g., methadone, buprenorphine for opioid addiction).
5. **Support Groups:** Participation in groups like Narcotics Anonymous (NA) for ongoing support and accountability.
6. **Aftercare Programs:** Continuing care and support after the initial treatment to prevent relapse.

Strategies for Reducing Illicit Drug Use

1. **Policy and Legislation:** Implementing and enforcing laws to reduce the availability of illicit drugs.
2. **Healthcare Access:** Improving access to mental health and addiction services.
3. **Harm Reduction:** Providing services like needle exchange programs and supervised injection sites to reduce the harm associated with drug use.
4. **Family and Social Support:** Encouraging supportive relationships and providing resources for families affected by drug use.
5. **Research and Innovation:** Investing in research to develop new treatments and prevention strategies.

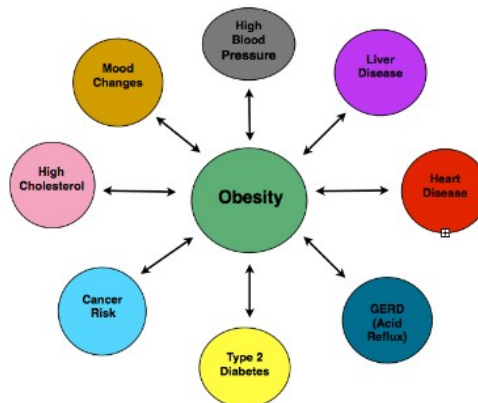
Outcome: The students learnt risk factors and various diseases caused by lifestyle.

1.3 Obesity

Objective: The students will be aware of Obesity and its effects.

Obesity is a medical condition characterized by an excessive accumulation of body fat that poses a risk to health. It is typically determined by body mass index (BMI), which is

calculated using a person's weight and height. A BMI of 30 or higher generally indicates obesity.



Several factors contribute to obesity, including:

1. **Unhealthy Diet:** Consuming high-calorie, low-nutrient foods such as fast food, sugary drinks, and processed snacks.
2. **Lack of Physical Activity:** Not engaging in enough physical activity to burn calories and maintain a healthy weight.
3. **Genetics:** Genetic factors can predispose individuals to obesity by influencing metabolism and fat storage.
4. **Behavioral Factors:** Poor sleep patterns, stress, and certain medications can also contribute to weight gain.

Obesity is associated with various health risks and complications, including type 2 diabetes, cardiovascular diseases (such as heart disease and stroke), certain cancers, sleep apnea, osteoarthritis, and psychological issues like depression. Preventing and managing obesity typically involves adopting a balanced diet, increasing physical activity, and making sustainable lifestyle changes. In some cases, medical interventions such as medications or surgery may be recommended, especially when other methods have not been effective. Obesity is a complex medical condition characterized by an excessive accumulation of body fat that presents a risk to health. It is typically assessed using body mass index (BMI), which is calculated by dividing a person's weight in kilograms by the square of their height in meters ($BMI = \text{weight (kg)} / \text{height}^2 (\text{m}^2)$). A BMI of 30 or higher generally indicates obesity.

Causes of Obesity:

1. **Unhealthy Diet:** Consuming foods high in calories, sugar, and fats, such as fast food, sugary drinks, and processed snacks, can contribute to weight gain.

2. **Lack of Physical Activity:** Not engaging in enough physical activity to burn calories and maintain a healthy weight can lead to obesity.
3. **Genetics:** Genetic factors can influence an individual's metabolism and fat storage, contributing to obesity.
4. **Environment:** Environmental factors such as easy access to unhealthy foods and sedentary lifestyles can promote weight gain.
5. **Psychological Factors:** Stress, lack of sleep, and certain mental health conditions can affect eating habits and contribute to obesity.

Health Risks and Complications:

Obesity is associated with numerous health risks and complications, including:

- **Type 2 Diabetes:** Obesity increases insulin resistance and the risk of developing type 2 diabetes.
- **Cardiovascular Diseases:** Excess weight can lead to high blood pressure, heart disease, and stroke.
- **Certain Cancers:** Obesity is linked to an increased risk of cancers such as breast, colon, and prostate cancers.
- **Respiratory Issues:** Obesity can cause breathing problems, including sleep apnea and asthma.
- **Joint Problems:** Excess weight puts stress on joints, leading to osteoarthritis and other joint-related issues.
- **Digestive Disorders:** Obesity increases the risk of gallbladder disease and fatty liver disease.
- **Psychological and Social Effects:** Obesity can impact self-esteem, mental health, and quality of life, leading to depression and social stigmatization.

Diagnosis and Treatment:

Diagnosis of obesity is primarily based on BMI, waist circumference, and assessment of other risk factors such as medical history and physical examinations. Treatment typically involves a combination of lifestyle changes, including:

- **Dietary Modifications:** Adopting a balanced diet rich in fruits, vegetables, lean proteins, and whole grains while limiting high-calorie and processed foods.
- **Physical Activity:** Increasing regular physical activity to burn calories and improve overall health.
- **Behavioral Therapy:** Counseling and behavior modification techniques can help individuals adopt healthier habits and manage stress.

- **Medications:** In some cases, medications may be prescribed to aid weight loss by reducing appetite or inhibiting fat absorption.
- **Surgery:** Bariatric surgery may be considered for severely obese individuals who have not responded to other treatments. Procedures such as gastric bypass or gastric sleeve surgery can help reduce food intake and promote weight loss.

Prevention:

Preventing obesity involves maintaining a healthy lifestyle from an early age, including:

- Eating a balanced diet and controlling portion sizes.
- Engaging in regular physical activity and limiting sedentary behaviors.
- Avoiding excessive consumption of high-calorie, low-nutrient foods and sugary drinks.
- Managing stress and getting enough quality sleep.
- Seeking regular medical check-ups and monitoring weight and BMI.

Conclusion:

Obesity is a significant public health challenge worldwide, affecting millions of people and contributing to a range of serious health conditions. Addressing obesity requires a multifaceted approach that includes individual behavior changes, community initiatives, and policy interventions aimed at promoting healthy lifestyles and reducing the prevalence of obesity-related diseases.

Outcome: The students learnt risk factors associated with obesity.

1.4 DIABETES

Objective: The students will be aware of diabetes and its types.

Diabetes, also known as diabetes mellitus, is a chronic medical condition that occurs when the body cannot properly regulate blood sugar (glucose) levels. Glucose is a crucial source of energy for cells and is derived from the foods we eat. Insulin, a hormone produced by the pancreas, helps glucose from the bloodstream enter cells to be used for energy.



Types of Diabetes:

1. Type 1 Diabetes:

- **Cause:** Type 1 diabetes is an autoimmune condition where the immune system mistakenly attacks and destroys insulin-producing cells (beta cells) in the pancreas. This results in little to no insulin production.
- **Onset:** Often diagnosed in children and young adults, although it can occur at any age.
- **Treatment:** Requires lifelong insulin therapy to manage blood sugar levels.

2. Type 2 Diabetes:

- **Cause:** Type 2 diabetes develops when the body becomes resistant to insulin or when the pancreas fails to produce enough insulin. It is often associated with lifestyle factors such as obesity, physical inactivity, and unhealthy diet.
- **Onset:** Typically diagnosed in adults, but increasingly seen in children and adolescents due to rising obesity rates.
- **Treatment:** Managed through lifestyle changes (diet, exercise) and medications (oral medications and/or insulin) to control blood sugar levels.

3. Gestational Diabetes:

- **Cause:** Gestational diabetes occurs during pregnancy when the body cannot produce enough insulin to meet the increased needs, resulting in high blood sugar levels.
- **Onset:** Develops during pregnancy and usually resolves after childbirth.
- **Treatment:** Managed through dietary changes and sometimes requires insulin therapy to maintain blood sugar levels within a normal range.

Symptoms of Diabetes:

- **Increased Thirst and Urination:** Excess glucose in the bloodstream can lead to frequent urination, which can cause dehydration and increased thirst.
- **Extreme Hunger:** Without enough insulin to move sugar into cells, muscles and organs become depleted of energy, leading to increased hunger.
- **Fatigue:** Lack of energy because glucose is not properly utilized by cells.
- **Blurry Vision:** High blood sugar levels can cause fluid to be pulled from tissues, including the lenses of the eyes, resulting in blurred vision.
- **Slow Healing:** High blood sugar levels can affect blood flow and impair the immune system, leading to slow wound healing.

Complications of Diabetes:

- **Cardiovascular Disease:** Diabetes increases the risk of heart disease, stroke, and narrowing of blood vessels (atherosclerosis).
- **Kidney Damage (Diabetic Nephropathy):** Diabetes can damage the kidneys' filtering system, leading to kidney failure.
- **Nerve Damage (Diabetic Neuropathy):** Diabetes can damage nerves throughout the body, leading to numbness, tingling, and pain, especially in the hands and feet.
- **Eye Damage (Diabetic Retinopathy):** Diabetes can damage the blood vessels in the retina, potentially leading to blindness.
- **Foot Damage:** Nerve damage and poor blood circulation in the feet increase the risk of infections, ulcers, and possibly amputation.

Management and Prevention:

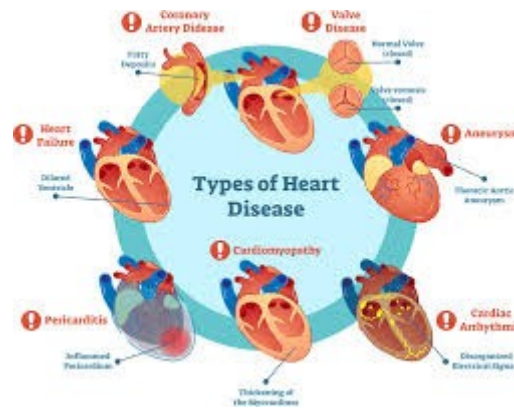
- **Blood Sugar Monitoring:** Regular monitoring of blood sugar levels is essential to manage diabetes effectively.
- **Healthy Eating:** Following a balanced diet that includes fruits, vegetables, whole grains, lean proteins, and healthy fats while limiting sugary foods and refined carbohydrates.
- **Physical Activity:** Engaging in regular physical activity helps control weight, lower blood sugar levels, and improve overall health.
- **Medications:** Depending on the type and severity of diabetes, medications such as insulin, oral medications, or other injectable medications may be prescribed.
- **Lifestyle Modifications:** Quitting smoking, managing stress, and getting regular medical check-ups are important for overall diabetes management.

Outcome: The students learnt how to avoid diabetics and prevention measures.

1.5 Cardiovascular diseases (CVD)

Objective: The students can know their heart related diseases.

Cardiovascular diseases (CVD) are a group of disorders affecting the heart and blood vessels. These diseases involve conditions that can affect the heart (cardio) and/or the blood vessels (vascular), leading to various complications. Cardiovascular diseases are a significant cause of morbidity and mortality worldwide.



Types of Cardiovascular Diseases:

1. Coronary Artery Disease (CAD):

- **Cause:** CAD occurs when the coronary arteries (the blood vessels that supply blood to the heart muscle) become narrowed or blocked due to a buildup of cholesterol and other substances (plaque) on the artery walls.
- **Complications:** Can lead to chest pain (angina), heart attack (myocardial infarction), and heart failure.

2. Hypertension (High Blood Pressure):

- **Cause:** Hypertension occurs when the force of blood against the walls of the arteries is consistently too high.
- **Complications:** Increases the risk of heart disease, stroke, and other cardiovascular complications.

3. Heart Failure:

- **Cause:** Heart failure happens when the heart muscle becomes weakened or stiff, leading to inadequate pumping of blood to meet the body's needs.
- **Complications:** Results in fatigue, shortness of breath, fluid buildup (edema), and increased risk of other heart-related problems.

4. Stroke:

- **Cause:** Stroke occurs when blood flow to part of the brain is interrupted or reduced, depriving brain tissue of oxygen and nutrients.
- **Types:** Ischemic stroke (caused by a blockage in a blood vessel) and hemorrhagic stroke (caused by a burst blood vessel).
- **Complications:** Can cause permanent brain damage, disability, or death.

5. **Peripheral Artery Disease (PAD):**

- **Cause:** PAD occurs when narrowed arteries reduce blood flow to the limbs, usually the legs.
- **Complications:** Leads to pain, numbness, tissue damage, and increases the risk of infections and amputations.

Risk Factors for Cardiovascular Diseases:

- **Unhealthy Diet:** High intake of saturated fats, trans fats, salt, and sugar can contribute to the development of CVD.
- **Physical Inactivity:** Lack of regular physical activity increases the risk of obesity, high blood pressure, and other risk factors for CVD.
- **Tobacco Use:** Smoking and exposure to secondhand smoke are major risk factors for CVD.
- **High Blood Pressure:** Hypertension strains the heart and blood vessels over time.
- **High Blood Cholesterol:** Elevated levels of cholesterol can contribute to plaque buildup in arteries.
- **Diabetes:** Increases the risk of CVD due to its effects on blood vessels and metabolism.
- **Family History:** Having close relatives with CVD increases an individual's risk.
- **Age and Gender:** Risk increases with age and tends to be higher in men, although the risk in women rises after menopause.

Prevention and Management:

- **Healthy Lifestyle:** Eating a balanced diet rich in fruits, vegetables, whole grains, and lean proteins. Limiting saturated fats, trans fats, salt, and sugar. Engaging in regular physical activity and avoiding tobacco use.
- **Medications:** Depending on the specific condition, medications may be prescribed to control blood pressure, cholesterol levels, and manage other risk factors.
- **Medical Monitoring:** Regular health check-ups and screenings to monitor blood pressure, cholesterol levels, and overall cardiovascular health.

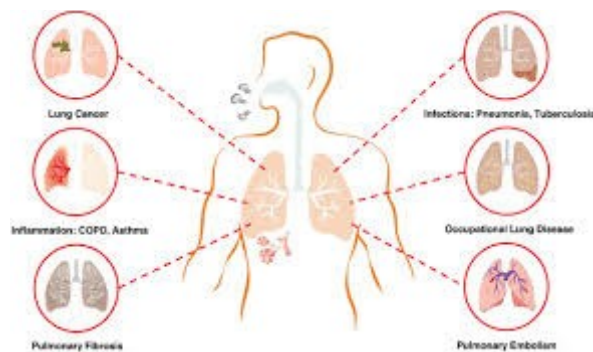
- **Stress Management:** Practicing stress-reducing techniques such as meditation, yoga, or hobbies can help lower the risk of CVD.

Outcome: The students learnt how to prevent and protect their heart from diseases.

1.6 Respiratory diseases

Objective: The students will learn the various Respiratory diseases.

Respiratory diseases encompass a broad range of conditions that affect the lungs and respiratory system. These diseases can vary in severity, from mild illnesses like the common cold to chronic conditions such as asthma or chronic obstructive pulmonary disease (COPD), and even life-threatening diseases like pneumonia or lung cancer.



Common Respiratory Diseases:

1. Asthma:

- **Cause:** Asthma is a chronic condition characterized by inflammation and narrowing of the airways, making breathing difficult.
- **Symptoms:** Wheezing, shortness of breath, chest tightness, and coughing, particularly at night or early morning.
- **Management:** Inhalers (bronchodilators and anti-inflammatory medications), avoiding triggers (e.g., allergens, pollutants), and monitoring symptoms.

2. Chronic Obstructive Pulmonary Disease (COPD):

- **Cause:** COPD is a progressive lung disease characterized by airflow limitation due to chronic bronchitis (inflammation of the bronchial tubes) and/or emphysema (damage to the lung tissue).
- **Symptoms:** Persistent cough, sputum production, shortness of breath, and wheezing.
- **Management:** Smoking cessation, medications (bronchodilators, steroids), pulmonary rehabilitation, and oxygen therapy in severe cases.

3. Pneumonia:

- **Cause:** Pneumonia is an infection of the lungs caused by bacteria, viruses, fungi, or parasites.
- **Symptoms:** Fever, cough, chest pain, rapid breathing, and difficulty breathing.
- **Treatment:** Antibiotics (for bacterial pneumonia), antiviral or antifungal medications, supportive care (e.g., oxygen therapy), and vaccines for prevention (e.g., pneumococcal vaccine, influenza vaccine).

4. Influenza (Flu):

- **Cause:** Influenza is a viral infection affecting the respiratory system caused by influenza viruses.
- **Symptoms:** Sudden onset of fever, chills, sore throat, cough, body aches, and fatigue.
- **Treatment:** Antiviral medications (if started early), supportive care, and annual vaccination for prevention.

5. Interstitial Lung Diseases (ILDs):

- **Cause:** ILDs encompass a group of disorders characterized by inflammation and scarring (fibrosis) of the lung tissue between the air sacs (alveoli).
- **Examples:** Idiopathic pulmonary fibrosis (IPF), sarcoidosis, and hypersensitivity pneumonitis.
- **Management:** Treatment depends on the specific ILD and may include medications (immunosuppressants, antifibrotic drugs), oxygen therapy, and pulmonary rehabilitation.

6. Lung Cancer:

- **Cause:** Lung cancer develops when abnormal cells in the lungs grow uncontrollably, usually due to exposure to carcinogens such as tobacco smoke, radon, or asbestos.
- **Symptoms:** Persistent cough, chest pain, shortness of breath, coughing up blood, and unexplained weight loss.
- **Treatment:** Depends on the type and stage of cancer and may include surgery, chemotherapy, radiation therapy, targeted therapy, and immunotherapy.

Prevention and Management:

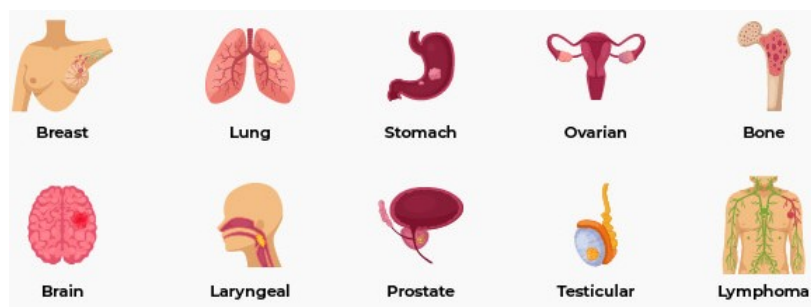
- **Smoking Cessation:** Quitting smoking is the most important preventive measure for respiratory diseases like COPD and lung cancer.
- **Vaccination:** Annual influenza vaccination and pneumococcal vaccination (especially for older adults and those at risk) can prevent infections that lead to respiratory complications.
- **Avoiding Exposure:** Minimizing exposure to environmental pollutants, allergens, and occupational hazards (e.g., asbestos, silica).
- **Healthy Lifestyle:** Eating a balanced diet, exercising regularly, maintaining a healthy weight, and managing chronic conditions like asthma or allergies effectively.
- **Regular Medical Check-ups:** Monitoring respiratory health through regular check-ups, especially for individuals with chronic respiratory conditions or risk factors.

Outcome: The students learnt the importance of Lungs and healthcare.

1.7 CANCER

Objective: The students get awareness of cancer and its effects.

Cancer is a complex group of diseases characterized by the uncontrolled growth and spread of abnormal cells. These cells can invade and destroy healthy tissues and organs, leading to serious health complications. Cancer can develop almost anywhere in the body and can affect people of all ages, although the risk increases with age.



Types of Cancer:

1. **Carcinomas:** Cancer that begins in the skin or tissues that line or cover internal organs. Examples include breast, lung, prostate, and colorectal cancers.
2. **Sarcomas:** Cancer that begins in the bones, cartilage, fat, muscle, blood vessels, or other connective or supportive tissues.

3. **Leukemias:** Cancer that starts in blood-forming tissue such as the bone marrow and causes large numbers of abnormal blood cells to be produced and enter the bloodstream.
4. **Lymphomas:** Cancer that begins in the lymphatic system, which is part of the body's immune system. It includes Hodgkin lymphoma and non-Hodgkin lymphoma.
5. **Central Nervous System (CNS) Cancers:** Cancer that begins in the tissues of the brain and spinal cord.

Causes of Cancer:

- **Genetic Factors:** Inherited mutations (changes) in certain genes can increase the risk of developing cancer. However, most cancers are not inherited but result from acquired mutations during a person's lifetime.
- **Environmental Factors:** Exposure to carcinogens (substances that can cause cancer) such as tobacco smoke, ultraviolet (UV) radiation from the sun, ionizing radiation (e.g., from medical imaging or nuclear fallout), and certain chemicals (e.g., asbestos, benzene) can contribute to cancer development.
- **Lifestyle Factors:** Unhealthy habits such as smoking, excessive alcohol consumption, lack of physical activity, and poor diet (high in processed foods and low in fruits and vegetables) can increase the risk of developing cancer.

Symptoms of Cancer:

- Persistent Cough or Hoarseness
- Changes in Bowel or Bladder Habits
- Sores That Do Not Heal
- Unexplained Weight Loss
- Fatigue
- Difficulty Swallowing
- Changes in Skin Moles or Wounds

Diagnosis and Treatment:

- **Diagnosis:** Cancer diagnosis usually involves a combination of imaging tests (e.g., X-rays, CT scans), laboratory tests (e.g., blood tests, biopsy), and sometimes genetic testing to determine the type and extent (stage) of cancer.

- **Treatment:** Treatment options depend on the type of cancer, its stage, and other individual factors. Common treatments include surgery, chemotherapy, radiation therapy, targeted therapy, immunotherapy, hormone therapy, and bone marrow or stem cell transplantation.

Cancer prevention

Cancer prevention involves reducing the risk of developing cancer through lifestyle changes, vaccination, and screening. While not all cancers are preventable, there are several strategies that can significantly lower the risk of certain types of cancer:

Lifestyle Factors:

1. Tobacco Use:

- **Risk:** Tobacco use is linked to many types of cancer, including lung, throat, mouth, pancreas, bladder, and cervix.
- **Prevention:** Quitting smoking and avoiding secondhand smoke significantly reduce cancer risk. Support programs and medications are available to help quit smoking.

2. Healthy Diet:

- **Risk:** Poor diet, high in processed foods, red meats, and low in fruits and vegetables, is associated with increased cancer risk.
- **Prevention:** Eating a balanced diet rich in fruits, vegetables, whole grains, and lean proteins can lower the risk of developing various cancers. Limiting intake of processed and red meats, and avoiding sugary drinks can also help.

3. Physical Activity:

- **Risk:** Lack of regular physical activity is associated with an increased risk of cancer, including breast, colon, and prostate cancers.
- **Prevention:** Engaging in regular physical activity (at least 150 minutes of moderate-intensity exercise per week) can help maintain a healthy weight and reduce cancer risk.

4. Limiting Alcohol Consumption:

- **Risk:** Alcohol consumption is linked to an increased risk of cancers of the mouth, throat, esophagus, liver, breast, and colon.
- **Prevention:** Limiting alcohol intake or avoiding it altogether can reduce cancer risk. For those who choose to drink, it is recommended to do so in moderation (up to one drink per day for women and up to two drinks per day for men).

5. UV Protection:

- **Risk:** Exposure to ultraviolet (UV) radiation from the sun and tanning beds increases the risk of skin cancer, including melanoma.
- **Prevention:** Using sunscreen with SPF 30 or higher, wearing protective clothing (wide-brimmed hats, sunglasses), seeking shade, and avoiding indoor tanning can help protect against skin cancer.

Vaccination:

1. HPV Vaccine:

- **Risk:** Human papillomavirus (HPV) infection is a risk factor for cervical, anal, and oropharyngeal cancers.
- **Prevention:** Vaccination against HPV is recommended for adolescents and young adults to reduce the risk of HPV-related cancers.

2. Hepatitis B Vaccine:

- **Risk:** Chronic hepatitis B infection increases the risk of liver cancer.
- **Prevention:** Vaccination against hepatitis B is recommended to prevent hepatitis B infection and reduce the risk of liver cancer.

Screening:

1. Cervical Cancer Screening:

- **Screening Test:** Pap smear (Pap test) and HPV testing.
- **Age Recommendations:** Starting at age 21 (Pap test) and 30 (HPV test) and continuing at regular intervals as recommended by healthcare providers.

2. Breast Cancer Screening:

- **Screening Test:** Mammography.
- **Age Recommendations:** Starting at age 40 or earlier if at higher risk, and continuing at regular intervals as recommended.

3. Colorectal Cancer Screening:

- **Screening Tests:** Colonoscopy, sigmoidoscopy, fecal occult blood test (FOBT), fecal immunochemical test (FIT).
- **Age Recommendations:** Starting at age 45 for average-risk individuals and earlier for those with risk factors or family history.

Outcome: The students learnt the cause and effects of cancer.

1.8 Diet and exercise

Objective: The students get familiarized with diet and exercise.

Diet plays a crucial role in overall health and well-being, influencing various aspects of physical and mental health, energy levels, and disease prevention. Here's an overview of key aspects related to diet:



Diet:

1. Nutrient-Dense Foods:

- **Benefits:** A balanced diet rich in fruits, vegetables, whole grains, lean proteins, and healthy fats provides essential nutrients like vitamins, minerals, antioxidants, and fiber.
- **Impact:** Supports overall health, boosts immune function, and reduces the risk of chronic diseases such as heart disease, diabetes, and certain cancers.

2. Portion Control and Moderation:

- **Benefits:** Monitoring portion sizes helps manage caloric intake, preventing overeating and promoting weight management.
- **Impact:** Helps maintain a healthy weight and reduces the risk of obesity-related conditions.

3. Hydration:

- **Benefits:** Drinking an adequate amount of water supports bodily functions, aids digestion, and helps regulate body temperature.
- **Impact:** Promotes overall health and well-being, enhances physical performance during exercise, and supports metabolism.

4. Nutrition Timing:

- **Benefits:** Eating balanced meals and snacks throughout the day provides sustained energy levels and supports muscle recovery after exercise.
- **Impact:** Optimizes performance during workouts and aids in muscle repair and growth.

Exercise:

1. Aerobic Exercise:

- **Benefits:** Activities like brisk walking; running, cycling, and swimming improve cardiovascular health, increase endurance, and burn calories.
- **Impact:** Helps maintain a healthy weight, reduces the risk of heart disease, and improves overall fitness levels.

2. Strength Training:

- **Benefits:** Involves resistance exercises using weights, resistance bands, or body weight to build muscle strength, tone muscles, and increase metabolism.
- **Impact:** Supports bone health, enhances physical function, and helps maintain muscle mass, especially important as we age.

3. Flexibility and Balance Exercises:

- **Benefits:** Activities like yoga, Pilates, and tai chi improve flexibility, balance, and coordination.
- **Impact:** Reduces the risk of falls, improves posture, and enhances overall mobility and range of motion.

Synergistic Effects of Diet and Exercise:

1. Weight Management:

- **Combined Impact:** A balanced diet helps control caloric intake, while regular exercise burns calories and builds muscle, promoting weight loss or maintenance.

2. Cardiovascular Health:

- **Combined Impact:** Eating a heart-healthy diet (low in saturated fats and sodium) and engaging in aerobic exercise improve heart function, reduce blood pressure, and lower cholesterol levels.

3. Energy Levels and Mood:

- **Combined Impact:** Nutrient-dense foods provide sustained energy, while exercise releases endorphins ("feel-good" hormones), reducing stress and improving mood.

4. Long-term Health Benefits:

- **Combined Impact:** Adopting a healthy diet and regular exercise routine lowers the risk of chronic diseases such as type 2 diabetes, osteoporosis, and certain cancers.

Practical Tips:

- **Set Realistic Goals:** Start with small, achievable changes in diet and exercise habits.
- **Plan and Prepare:** Schedule workouts and meals in advance, ensuring you have nutritious options readily available.
- **Stay Consistent:** Establish a routine that incorporates both diet and exercise, aiming for regular physical activity and balanced meals throughout the week.

Components of a Healthy Diet:**1. Fruits and Vegetables:**

- **Benefits:** Rich in vitamins, minerals, antioxidants, and dietary fiber, which support overall health and reduce the risk of chronic diseases such as heart disease, stroke, and certain cancers.
- **Recommendation:** Aim for a variety of colors (e.g., dark leafy greens, orange carrots, red peppers) and include them in meals and snacks daily.

2. Whole Grains:

- **Benefits:** Provide essential nutrients like fiber, B vitamins, and minerals. Whole grains promote digestive health, help regulate blood sugar levels, and reduce the risk of heart disease.
- **Examples:** Whole wheat, brown rice, oats, quinoa, barley, and whole grain pasta.

3. Proteins:

- **Benefits:** Important for building and repairing tissues, producing enzymes and hormones, and maintaining immune function.
- **Sources:** Lean meats (like chicken, turkey, lean cuts of beef and pork), fish, eggs, legumes (beans and lentils), nuts, seeds, and tofu.

4. Healthy Fats:

- **Benefits:** Essential for brain health, hormone production, and absorption of fat-soluble vitamins (A, D, E, K). Monounsaturated and polyunsaturated fats can lower bad cholesterol levels (LDL) and reduce the risk of heart disease.
- **Sources:** Olive oil, avocados, nuts (like almonds, walnuts), seeds (like chia seeds, flaxseeds), and fatty fish (such as salmon, trout, sardines).

5. Dairy or Dairy Alternatives:

- **Benefits:** Good sources of calcium, vitamin D, protein, and other essential nutrients for bone health and overall well-being.

- **Options:** Milk, yogurt, cheese, and fortified plant-based alternatives (like almond milk, soy milk).

Tips for a Healthy Diet:

- **Balance and Variety:** Include a variety of foods from different food groups to ensure you get a wide range of nutrients.
- **Portion Control:** Pay attention to portion sizes to avoid overeating, which can contribute to weight gain.
- **Limit Added Sugars and Salt:** Minimize intake of sugary beverages, candies, processed foods, and high-sodium foods to reduce the risk of obesity, diabetes, and hypertension.
- **Stay Hydrated:** Drink plenty of water throughout the day to maintain hydration and support bodily functions.

Special Considerations:

- **Weight Management:** A balanced diet combined with regular physical activity is key to achieving and maintaining a healthy weight.
- **Medical Conditions:** Certain conditions (like diabetes, high blood pressure, or food allergies) may require specific dietary modifications. Consult with a healthcare professional or registered dietitian for personalized advice.
- **Cultural and Personal Preferences:** Dietary choices can vary based on cultural background, ethical beliefs (e.g., vegetarianism or veganism), or personal preferences. It's important to adapt dietary guidelines to individual needs and lifestyles.

Outcome: The students learnt the different types of diet and exercise for good health.



UNIT 2 OBT356 - unit notes

Food ,Nutrition and health (Anna University)



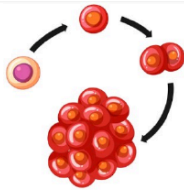
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UNIT II CANCER

Types - Lung cancer, Mouth cancer, Skin cancer, Cervical cancer, Carcinoma esophagus; Causes Tobacco usage, Diagnosis – Biomarkers, Treatment

Cancer:

Cancer is a **disease in which some of the body's cells grow uncontrollably and** spread to other parts of the body.



Cancer cells are cells that undergo uncontrolled growth and division, resulting in the development of an abnormal tissue mass referred to as a tumor.

Causes:

- Alcohol
- Tobacco
- Smoking
- Hormones
- Chemicals
- Immunosuppression
- Infection
- Obesity
- Radiation
- Chronic inflammation
- Genetic
- Asbestos
- Radiation
- Ultraviolet
- Physical inactivity

Symptoms:

- Fatigue
- Lump or area of thickening that can be felt under the skin
- Weight changes, including unintended loss or gain
- Skin changes, such as yellowing, darkening or redness of the skin, sores that won't heal, or changes to existing moles

- Changes in bowel or bladder habits
- Persistent cough or trouble breathing
- Difficulty swallowing
- Hoarseness
- Persistent indigestion or discomfort after eating
- Persistent, unexplained muscle or joint pain
- Persistent, unexplained fevers or night sweats
- Unexplained bleeding or bruising

Prevention:

Cancer risk can be reduced by:

- not using tobacco;
- maintaining a healthy body weight;
- eating a healthy diet, including fruit and vegetables;
- doing physical activity on a regular basis;
- avoiding or reducing consumption of alcohol;
- getting vaccinated against HPV and hepatitis B if you belong to a group for which vaccination is recommended;
- avoiding ultraviolet radiation exposure (which primarily results from exposure to the sun and artificial tanning devices) and/or using sun protection measures;
- ensuring safe and appropriate use of radiation in health care (for diagnostic and therapeutic purposes);
- minimizing occupational exposure to ionizing radiation; and
- Reducing exposure to outdoor air pollution and indoor air pollution, including radon (a radioactive gas produced from the natural decay of uranium, which can accumulate in buildings — homes, schools and workplaces).

Types:

There are more than 100 types of cancer.

Some common types are

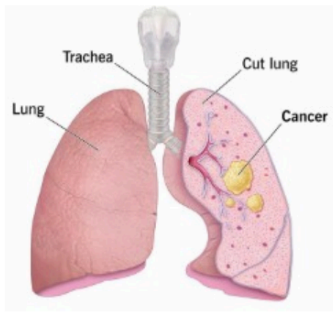
- Blood & Lymphoid Cancer. Hodgkin Lymphoma. Leukemia.
- Bone & Soft Tissue Cancer.
- Brain & Central Nervous System Cancer.
- Breast Cancer.
- Digestive System Cancer.
- Head & Neck Cancer.
- Lung Cancer. Lung. Mesothelioma.
- Pelvic Area Cancer. Gestational Trophoblastic Neoplasia. Prostate.
- Urinary Cancer

- Skin cancer
 - Eye cancer
-
-

Lung cancer:

Lung cancer is a kind of cancer that starts as a growth of cells in the lungs. The lungs are two spongy organs in the chest that control breathing.

People who smoke have the greatest risk of lung cancer. The risk of lung cancer increases with the length of time and number of cigarettes smoked. Quitting smoking, even after smoking for many years, significantly lowers the chances of developing lung cancer. Lung cancer also can happen in people who have never smoked.



Symptoms

Lung cancer typically doesn't cause symptoms early on. Symptoms of lung cancer usually happen when the disease is advanced.

Signs and symptoms of lung cancer that happen in and around the lungs may include:

- A new cough that doesn't go away.
- Chest pain.
- Coughing up blood, even a small amount.
- Hoarseness.
- Shortness of breath.
- Wheezing.

Signs and symptoms that happen when lung cancer spreads to other parts of the body may include:

- Bone pain.

- Headache.
- Losing weight without trying.
- Loss of appetite.
- Swelling in the face or neck.

Types of lung cancer

Lung cancer is divided into two major types based on the appearance of the cells under a microscope. The healthcare professional makes treatment decisions based on which major type of lung cancer. The two general types of lung cancer include:

- **Small cell lung cancer.** Small cell lung cancer usually only happens in people who have smoked heavily for years. Small cell lung cancer is less common than non-small cell lung cancer.
-
- **Non-small cell lung cancer.** Non-small cell lung cancer is a category that includes several types of lung cancers. Non-small cell lung cancers include squamous cell carcinoma, adenocarcinoma and large cell carcinoma.
-

Risk factors:

A number of factors may increase the risk of lung cancer. Some risk factors can be controlled, for instance, by quitting smoking. Other factors can't be controlled, such as your family history.

Risk factors for lung cancer include:

- Smoking
- Exposure to secondhand smoke
- Previous radiation therapy
- Exposure to radon gas
- Exposure to cancer-causing substances
- Family history of lung cancer

Complications

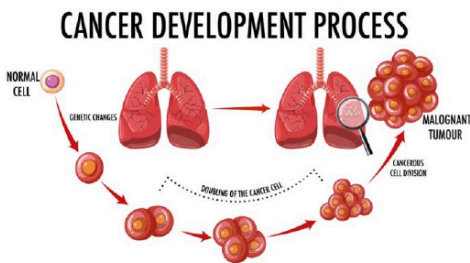
- Lung cancer can cause complications, such as:
- Shortness of breath
- Coughing up blood
- Pain
- Fluid in the chest
- Cancer that spreads to other parts of the body

Prevention

There's no sure way to prevent lung cancer, but you can reduce your risk if you:

- Don't smoke
- Stop smoking
- Avoid secondhand smoke
- Test your home for radon
- Avoid carcinogens at work
- Eat a diet full of fruits and vegetables
- Exercise most days of the week
- If you don't exercise regularly, start out slowly. Try to exercise most days of the week.

Lung Cancer Development process:



Stages of Lung Cancer

Stage 0 (in-situ): Cancer is present in the top lining of the lung or bronchus, without spreading to other lung parts or outside the lung itself.

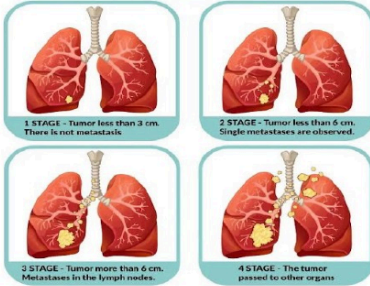
Stage I: Cancer has not spread outside the lung.

Stage II: Cancer is larger than Stage I, has spread to nearby lymph nodes within the lung, or multiple tumors are present in the same lung lobe.

Stage III: Cancer is larger than Stage II, has spread to nearby lymph nodes or adjacent structures, or multiple tumors are found in a different lobe of the same lung.

Stage IV: Cancer has spread to the other lung, the fluid surrounding the lung, the fluid around the heart, or distant organs.

STAGE LUNGS CANCER



Diagnosis of Lung Cancer:

A number of tests can be used to look for cancerous cells and to rule out other conditions.

Tests may include:

- **Imaging tests.** Imaging tests make pictures of the body. They can show the location and size of the lung cancer. Tests might include X-ray, MRI, CT and positron emission tomography, which also is called a PETscan.
- **Sputum cytology.** Sputum is the mucus that is coughed up from the lungs. If coughing are in up sputum, it can be looked at under a microscope. The sputum can sometimes show lung cancer cells.
- **Biopsy.** A biopsy is a procedure to remove a sample of tissue for testing in a lab.

Healthcare team can perform a lung cancer biopsy several ways. One way is bronchoscopy. During bronchoscopy, a healthcare professional passes a lighted tube with a camera down the throat into the lungs to examine the area. Special tools can be passed through the tube to collect a sample of tissue.

Mediastinoscopy also is an option. During mediastinoscopy, an incision is made at the base of the neck. Surgical tools are then inserted behind the breastbone to take tissue samples from lymph nodes.

Another option is a needle biopsy. In a needle biopsy, the healthcare professional uses X-ray or CT images to guide a needle through the skin

on the chest. The needle goes into the lung tissue to collect cells that could be cancerous.

A biopsy sample also may be taken from lymph nodes or other areas where cancer has spread.

Cancer cells will be carefully tested in a lab to find out what type of lung cancer. The results can help determine the likely outcome of the cancer, called the prognosis, and guide the treatment.

Tests to determine the extent of the cancer

If a patient is diagnosed with lung cancer, he/she may have other tests to see if the cancer has spread. These tests help the healthcare team find out the extent of the cancer, also called the stage. Cancer staging tests often involve imaging tests. The tests might look for signs of cancer in the lymph nodes or in other parts of the body. The healthcare team uses the cancer staging test results to help create the treatment plan.

Imaging tests may include MRI, CT, bone scans and PET scan. Not every test is right for every person.

Treatment:

Treatments for lung cancer are designed to get rid of cancer in your body or slow down its growth. Treatments can remove cancerous cells, help to destroy them or keep them from multiplying or teach the immune system to fight them. Some therapies are also used to reduce symptoms and relieve pain. Treatment will depend on the type of lung cancer, where it is, how far it's spread and many other factors.

Lung cancer treatments include surgery, radiofrequency ablation, radiation therapy, chemotherapy, targeted drug therapy and immunotherapy

Surgery:

NSCLC that hasn't spread and SCLC that's limited to a single tumor can be eligible for surgery. The surgeon might remove the tumor and a small amount of healthy tissue around it to make sure they don't leave any cancer cells behind. Sometimes they have to remove all or part of the Lung for the best chance that cancer won't come back

Radio frequency ablation:

NSCLC tumors near the outer edges of the lungs are sometimes treated with radiofrequency ablation (RFA). RFA uses high-energy radio waves to heat and destroy cancer cells.

Radiation Therapy:

Radiation uses high energy beams to kill cancer cells. It can be used by itself or to help make surgery more effective. Radiation can also be used as

palliative care, to shrink tumors and relieve pain. It's used in both NSCLC and SCLC.

Chemotherapy

Chemotherapy is often a combination of multiple medications designed to stop cancer cells from growing. It can be given before or after surgery or in combination with other types of medication, like immunotherapy. Chemotherapy for lung cancer is usually given through an IV

Targeted Drug therapy

In some people with NSCLC, lung cancer cells have specific changes (mutations) that help the cancer grow. Special drugs target these mutations to try to slow down or destroy cancer cells. Other drugs, called angiogenesis inhibitors, can keep the tumor from creating new blood vessels, which the cancer cells need to grow

Immunotherapy

Our bodies usually recognize cells that are damaged or harmful and destroy them. Cancer has ways to hide from the immune system to keep from being destroyed. Immunotherapy reveals cancer cells to the immune system so that the body can fight cancer

Treatments to ease symptoms (palliative care)

Some lung cancer treatments are used to relieve symptoms, like pain and difficulty breathing. These include therapies to reduce or remove tumors that are blocking airways, and procedures to remove fluid from around your lungs and keep it from coming back

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MOUTH CANCER:

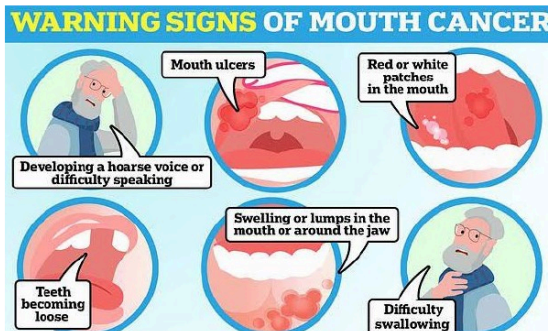
Mouth cancer, also known as oral cancer or cancer of the oral cavity, is often used to describe a number of cancers that start in the region of the mouth. These most commonly occur on the lips, tongue and floor of the mouth but can also start in the cheeks, gums, roof of the mouth, tonsils and salivary glands

Mouth cancer symptoms

Symptoms of mouth cancer can include:

- a lump in your neck
- loose teeth
- swelling or a sore on your lip that won't heal
- difficult or painful swallowing
- changes in speech
- bleeding or numbness in the mouth
- white or red patches on the mouth, tongue or gums
- unexplained weight loss.

Warning signs of mouth cancer:



Causes of mouth cancer

The main risk factors for most mouth cancers are tobacco, drugs and alcohol consumption. Other risk factors can include:

- human papillomavirus (HPV)
- Epstein-Barr virus (EBV)
- family history of mouth cancers
- poor oral hygiene and gum disease
- exposure to the sun
- chewing the seed of the areca palm tree (sometimes called areca or betel nut).

Diagnosis of mouth cancer

Signs of mouth cancer are often first detected by the dentist. Dentist examines the mouth, throat, tongue, cheeks, ears and eyes.

Endoscopy

In an endoscopy, a flexible tube (endoscope) is used to examine the nose, sinuses, larynx (voice box) and pharynx (throat).

Biopsy

A biopsy is the removal of a small amount of tissue which is sent to a pathologist to determine if cancer cells are present.

X-rays

X-rays of the head and neck can be used to see if cancer cells are present in the jaw, lungs or chest.

Scans

PET, CT and MRI scans to determine the location, stage and spread of cancer cells.

Treatment for mouth cancer

Treatment for mouth cancer depends on the type of cancer, where it is located and how far it has spread.

Staging

Tests such as CT and MRI scans help to determine how far the cancer has spread. Staging helps your doctor decide on the best treatment options for you.

Surgery

A common treatment for mouth cancers is surgery to remove the tumour, particularly in early-stage cancer. The extent of surgery may depend on the size of the tumour and how far the cancer has spread. Surrounding tissue and lymph nodes may also be removed.

Radiation therapy (radiotherapy)

Radiation therapy (also known as radiotherapy) can be used on its own to treat some small mouth cancers. Radiation therapy can also be used in combination with chemotherapy for more advanced cancers. Your doctor may also recommend radiation therapy after surgery.

Chemotherapy

Chemotherapy uses drugs to kill or shrink tumours. It can be used on its own or with radiation therapy. You may have chemotherapy to shrink a tumour before surgery or if cancer has come back after other treatments.

Palliative care

Palliative care aims to improve your quality of life without trying to cure the cancer. It may be recommended by your doctor to alleviate the symptoms of cancer or the side-effects of treatment.

SKIN CANCER:

Skin cancer is the uncontrolled growth of abnormal skin cells. It occurs when damage to skin cells—most often caused by ultraviolet radiation from sunshine or tanning beds—causes skin cells to multiply rapidly and form malignant tumors.

Most skin cancer is treatable if it's caught early

Symptoms:

The most common warning sign of skin cancer is a change on your skin — typically a new growth or a change in an existing growth or mole. Skin cancer symptoms include

- A new mole. Or a mole that changes in size, shape or color, or that bleeds
- A pearly or waxy bump on your face, ears or neck
- A flat, pink/red- or brown-colored patch or bump
- Areas on your skin that look like scars
- Sores that look crusty, have a depression in the middle or bleed often
- A wound or sore that won't heal, or that heals but comes back again
- A rough, scaly lesion that might itch, bleed and become crusty

Skin cancer looks different depending on the type of skin cancer

Thinking of the ABCDE rule tells what signs to watch for:

Asymmetry: Irregular shape

Border: Blurry or irregularly shaped edges

Color: Mole with more than one color

Diameter: Larger than a pencil eraser (6 millimeters)

Evolution: Enlarging, changing in shape, color or size

Causes:

The main cause of skin cancer is overexposure to sunlight especially when have sunburn and blistering. UV rays from the sun damage DNA in of the skin, causing abnormal cells to form. These abnormal cells rapidly divide in a disorganized way, forming a mass of cancer cells.

Risk Factors:

Anyone can get skin cancer, regardless of race or sex. But some groups get it more than others. Before the age of 50, skin cancer is more common in women. And it's about 30 times more common in white people than Black people or people of Asian/Pacific Islander descent. Unfortunately, skin cancer is often diagnosed in later stages for people with darker skin tone. Hence it makes more difficult to treat.

- Spend a considerable amount of time working or playing in the sun
- Get easily sunburned
- Live in a sunny or high-altitude climate
- Have light-colored eyes, blond or red hair and fair
- Have many moles or irregularly shaped moles
- Have actinic ketatosis (precancerous skin growths that are rough, scaly, dark pink-to-brown patches)
- Have a family history of skin cancer
- Have had an organ transplant
- Take medications that suppress or weaken your immune system

- Have been exposed to UV light therapy for treating skin conditions such as eczema or

Diagnosis:

Biopsy:

In a biopsy, a sample of tissue is removed and sent to a laboratory where a pathologist examines it under a microscope

Types:

Melanoma

Melanoma is often called "the most serious skin cancer" because it has a tendency to spread.

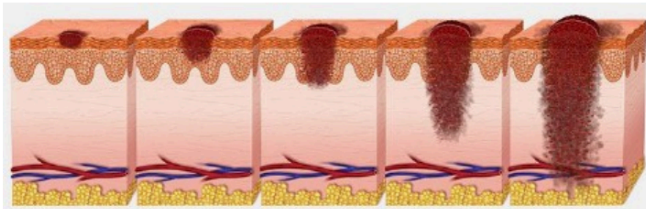
Non-melanoma

Non-melanoma skin cancer is a common type of cancer that starts in the top layer of skin. The main types are basal cell carcinoma (BCC) and squamous cell carcinoma (SCC). Non-melanoma skin cancer can often be easily treated. The main cause is ultraviolet light, which comes from the sun and is used in sunbeds.

Skin cancer stages:

Cancer stages tell you how much cancer is in your body. The stages of skin cancer range from stage 0 to stage IV. In general, the higher the number, the more cancer has spread and the harder it is to treat. But the staging for melanoma is different from non-melanoma skin cancers that start in your basal or squamous cells

Melanoma Staging:



Stage 0 (melanoma in situ): The melanoma is only in the top layer of your skin

Stage I: The melanoma is low risk and there's no evidence that it has spread. It's generally curable with surgery

Stage II: It has some features that indicate that it's likely to come back (recur), but there's no evidence of spread

Stage III: The melanoma has spread to nearby lymph nodes or nearby skin

Stage IV: The melanoma has spread to more distant lymph nodes or skin, or has spread to internal organs

Non Melanoma Staging:

Stage 0: Cancer is only in the top layer of your skin

Stage I (1): Cancer is in the top and middle layers of the skin

Stage II (2): Cancer is in the top and middle layers of the skin and moves to target the nerves or deeper layers of skin

Stage III (3): Cancer has spread beyond the skin to your lymph nodes

Stage IIIV (4): Cancer has spread to other parts of the body and the organs like liver, lungs or brain

Treatment:

Treatment depends on the stage of cancer. Sometimes, a biopsy alone can remove all the cancer tissue if it's small and limited to the surface of your skin. Other common skin cancer treatments, used alone or in combination, include

Radiation Therapy: The radiation oncologist uses radiation (strong beams of energy) to kill cancer cells or keep them from growing and dividing

Immunotherapy: The oncologist gives medications to train immune system to kill cancer cells

Chemotherapy: The dermatologist or Oncologist uses medications to kill cancer cells. Anticancer medications can be applied directly on the skin (topical chemotherapy) if limited to the skin's top layer or provided through pills or an IV if the cancer has spread to other parts of the body

Curettage and electrodesiccation: The dermatologist uses an instrument with a sharp, looped edge to remove cancer cells as it scrapes across the tumor. Then, they use an electric needle to destroy any remaining cancer cells. Providers often use this to treat basal cell and squamous cell cancers and precancerous skin tumors

Mohs Surgery: The dermatologist removes only diseased tissue, saving as much surrounding normal tissue as possible. Providers use this to treat basal cell and squamous cell cancers and, sometimes, other skin cancers that develop near sensitive or cosmetically important areas, like the eyelids, ears, lips, forehead, scalp, fingers or genital area

Excisional Surgery: The dermatologist removes the tumor and some surrounding healthy skin to be sure all the cancer is gone

Cryotherapy: The dermatologist uses liquid nitrogen to freeze skin cancer. The dead cells slough off after treatment

Photodynamic Therapy: The dermatologist coats skin with medication, which they activate with a blue or red fluorescent light. This therapy destroys precancerous cells while leaving normal cells alone

Prevention:

- Use a sunscreen with a skin protection factor (SPF) of 30 or higher. Broad-spectrum sunscreens protect against both UV-B and UV-A rays. Apply the sunscreen 30 minutes before you go outside. Wear sunscreen every day, even on cloudy days and during the winter months
 - Wear hats with wide brims to protect your face and ears
 - Wear long-sleeved shirts and pants to protect your arms and legs
 - Wear sunglasses to protect your eyes
 - Use a lip balm with sunscreen
 - Avoid the sun between 10 a.m. and 4 p.m.
 - Avoid tanning beds. If you want a tanned look, use a spray on tanning product
 - Use medication if skin more sensitive to sunlight
 - Regularly check all your skin for any changes in size, shape or color of skin growths or the development of new skin spots.
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CERVICAL CANCER

Cervical cancer is a growth of cells that starts in the cervix. The cervix is the lower part of the uterus that connects to the vagina

HPV infection causes cervical cancer

Long-lasting (persistent) infection with high-risk types of human papillomavirus (HPV) causes virtually all cervical cancers. Two high-risk types, HPV 16 and HPV 18, cause 70% of cervical cancers worldwide.

Nearly all people who are sexually active will become infected with HPV at some point in their lives. Most HPV infections go away on their own within a year or two as the immune system controls the infection. These short-term infections do not cause cancer. When a high-risk HPV infection lasts for years, it can lead to changes in the cervical cells, resulting in a precancerous

lesion. If the precancerous lesion is not found and removed, it may eventually develop into cervical cancer.

People who become sexually active at a young age, especially before age 18, or have multiple sexual partners are more likely to become infected with a high-risk type of HPV.

Factors that increase the risk that an HPV infection will cause cancer

Some risk factors make it more likely for a person who has a high-risk HPV infection of the cervix to develop cervical cancer. These risk factors include:

- **Having a weakened immune system.** This can lower the body's ability to fight an HPV infection. HPV infections are more likely to be persistent and progress to cancer in people who are immuno compromised than in people who are not immune compromised. You may be immune compromised if you:
 - have an HIV infection or another disease that weakens your immune system
 - take medicine to suppress your immune response, such as to prevent organ rejection after a transplant, to treat an autoimmune disease, or to treat cancer
- **Smoking or breathing in secondhand smoke.** People who smoke or breathe in secondhand smoke have an increased risk of developing cervical cancer. The risk increases the more a person smokes or is exposed to secondhand smoke.
- **Reproductive factors.** Both the use of oral contraceptives (birth control pills) and giving birth to many children are associated with an increased risk of cervical cancer. The reasons for these associations are not well understood.
- **Obesity.** Cervical cancer screening may be more difficult in those with obesity, leading to lower detection of precancers and a higher risk of cancer.

Cervical cancer is preventable

Cervical cancer is highly preventable and highly curable if caught early. Nearly all cervical cancers could be prevented by HPV vaccination, routine

cervical cancer screening, and appropriate follow-up treatment when needed.

HPV vaccination

HPV vaccination is a safe and effective way to help prevent cervical cancer.

Signs and Symptoms:

- Vaginal bleeding after intercourse, between periods or after menopause.
- Menstrual bleeding that is heavier and lasts longer than usual.
- Watery, bloody vaginal discharge that may be heavy and have a foul odor.
- Pelvic pain or pain during intercourse

Types:

- Cervical cancer is divided into types based on the type of cell in which the cancer begins. The main types of cervical cancer are:
 - **Squamous cell carcinoma.** This type of cervical cancer begins in thin, flat cells, called squamous cells. The squamous cells line the outer part of the cervix. Most cervical cancers are squamous cell carcinomas.
 - **Adenocarcinoma.** This type of cervical cancer begins in the column-shaped gland cells that line the cervical canal.

Treatments:

Surgery

Surgery (also called an operation) is sometimes used to treat cervical cancer. The type of surgery depends on where the cancer is located.

The following surgical procedures may be used:

Cold knife conization:

Cold knife conization uses a scalpel to remove a cone-shaped piece of tissue from the cervix and cervical canal. Sometimes all the cancer can be removed during this procedure. Cold knife conization is done in the hospital under general anesthesia.

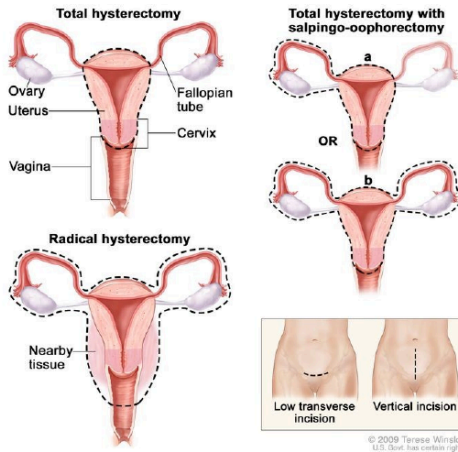
Sentinel lymph node biopsy:

Sentinel lymph node biopsy removes the sentinel lymph node during surgery. The sentinel lymph node is the first lymph node in a group of lymph nodes to receive lymphatic drainage from the primary tumor. It is therefore the first lymph node the cancer is likely to spread to from the primary tumor.

Hysterectomy

A hysterectomy is surgery to remove the uterus. As a treatment for cervical cancer, the cervix, and sometimes the surrounding structures, are removed. Several types of hysterectomy may be used to treat cervical cancer:

- **Total hysterectomy** removes the uterus and the cervix.
- **Radical hysterectomy** removes the uterus, cervix, part of the vagina, and a wide area of ligaments and tissues around these organs. The ovaries, fallopian tubes, or nearby lymph nodes may also be removed.
- **Modified radical hysterectomy** removes the uterus, cervix, upper part of the vagina, and ligaments and tissues that closely surround these organs. This type of surgery removes fewer tissues and/or organs than radical hysterectomy. The ovaries, fallopian tubes, or nearby lymph nodes may also be removed.



Radical trachelectomy

Radical trachelectomy (also called radical cervicectomy) removes the cervix, nearby tissue, and the upper part of the vagina. Lymph nodes may also be removed. After the surgeon removes the cervix, they attach the uterus to the remaining part of the vagina. A special stitch or band is placed on the uterus (in a procedure called a cerclage) to help keep the uterus closed during pregnancy. If you have this surgery, you may still be able to become pregnant.

Bilateral salpingo-oophorectomy

Bilateral salpingo-oophorectomy removes both ovaries and both fallopian tubes. This is done when the cancer has spread to these organs.

Total pelvic exenteration

Total pelvic exenteration removes the lower colon, rectum, and bladder. The cervix, vagina, ovaries, and nearby lymph nodes are also removed. Artificial openings (stoma) are made for urine and stool to flow from the body to a collection bag. Plastic surgery may be needed to make an artificial vagina after this operation.

Radiation therapy

Radiation therapy uses high-energy x-rays or other types of radiation to kill cancer cells or keep them from growing by damaging their DNA.

Both external and internal radiation therapy are used to treat cervical cancer and may also be used as palliative therapy to relieve symptoms and improve quality of life in people with advanced cervical cancer.

External radiation therapy

External-beam radiation therapy uses a machine outside the body to send radiation toward the area of the body with cancer.

Internal radiation therapy

Internal radiation therapy uses a radioactive substance sealed in needles, seeds, wires, or catheters that are placed directly into or near the cancer.

Chemotherapy

Chemotherapy (also called chemofl uses drugs to stop the growth of cancer cells, either by killing the cells or by stopping them from dividing.

Targeted therapy

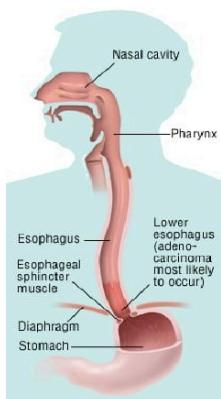
Targeted therapy uses drugs or other substances to block the action of specific enzymes, proteins, or other molecules involved in the growth and spread of cancer cells.

Immunotherapy

Immunotherapy helps a person's immune system fight cancer.

CARCINOMA ESOPHAGUS:

Cancer of the tube that runs from the throat to the stomach (esophagus).



Esophageal cancer is the abnormal growth of cells in the esophagus. The esophagus is the tube that carries food and liquid to your stomach.

Types:

- **Squamous cell carcinoma** starts in the cells that line the esophagus. These cells are called squamous cells. This type of cancer can occur anywhere in the esophagus.
- **Adenocarcinoma** starts in the lower portion of the esophagus, near the opening to the stomach. It begins when squamous cells are replaced by glandular cells that then begin growing abnormally.

Risk Factors

No one knows for sure what causes esophageal cancer. However, the most significant risk factors include the following:

- **Tobacco use** — The longer you smoke and the more you smoke each day, the greater your risk. Patients who develop esophageal cancer may also be at risk for developing other cancers of the head and neck.
- **Alcohol consumption** — Chronic or excessive consumption of alcohol, especially when combined with tobacco use, increases the risk.
- **Gastroesophageal reflux disease (GERD)**,— The lining of the stomach contains glandular cells that release acids and enzymes to digest food. Sometimes, these chemicals escape from the stomach and move into the esophagus. This is called reflux or GERD. One symptom of GERD is chronic heartburn.
- **Barrett's esophagus** — GERD is believed to irritate squamous cells near the stomach causing them to become glandular cells. This condition is called Barrett's esophagus.

Symptoms:

At first, esophageal cancer may not cause any symptoms. But as it progresses, it may cause

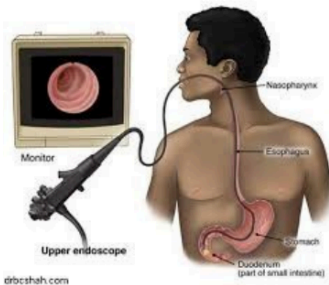
- trouble swallowing
- feeling like food is “stuck” in the chest
- pain in the chest or between the shoulder blades
- frequent heartburn or GERD
- severe weight loss
- hoarseness or chronic cough
- vomiting

Diagnosis

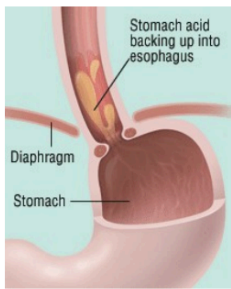
Routine laboratory studies — Basic blood tests can help determine if the patients have been losing blood and whether the organs are functioning normally. These tests can help the doctor to determine what other tests are needed.

Barium swallow — This test is an x-ray of the esophagus. Patient drinks a liquid containing barium, which coats the interior of your esophagus. It makes it easier for the doctor to see blockages or changes in the esophagus on x-rays.

Endoscopy — The doctor inserts a thin, lighted tube called an endoscope into the esophagus. A small video camera sits at the end of the tube. With this tool, the doctor can look for problems in the esophagus. He or she can also collect tissue samples from suspicious areas for examination. Patients will be given a sedative or pain medicine to minimize discomfort.



- **Computed tomography (CT) scan**,— X-ray pictures taken from different angles provide a three-dimensional view of the internal organs. Doctors can then see if they have any masses or blockages. CT scans are especially helpful in determining the extent of cancer. This information can guide treatment decisions.
- **Endoscopic ultrasound** — A tiny ultrasound machine sits at the end of a tube inserted into the esophagus. It creates pictures with sound waves. This test may be better than CT at determining how far cancer has grown into the esophagus, surrounding tissue, and lymph nodes.
- **PET scanning** — A positron emission tomography, or PET, scan is an imaging technique that uses positively charged particles (radioactive positrons) to detect subtle changes in the body's metabolism and chemical activities. A PET scan provides a color-coded image of the body's function, rather than its structure. Because the metabolic activity of cancer cells differs from normal cells, PET can detect cancer that has spread to other areas of the body. This information can affect your choice of treatment. People with esophageal squamous cell carcinoma have a higher risk of mouth, throat, lung, and stomach cancer.



Treatments:

Treatment Stages of esophageal cancer

Pathological stage (also called the *surgical stage*): If surgery is done first, the pathological stage is determined by examining tissue removed during an operation. This is the most common system used.

Clinical stage: If surgery might not be possible or will be done after other treatment is given, then the clinical stage is determined based on the results of a physical exam, biopsy, and imaging tests.

Postneoadjuvant stage: If chemotherapy or radiation is given before surgery (this is called *neoadjuvant therapy*), then a separate postneoadjuvant stage will be determined after surgery.

Treatment

Treatment for small esophageal cancers usually begins with surgery to remove the cancer. If the cancer grows larger or spreads to other parts of the body, treatment might start with chemotherapy and radiation instead.

Surgery:

Esophageal cancer surgery

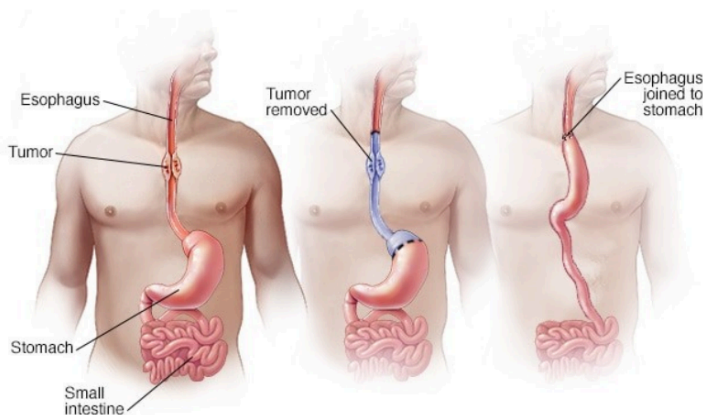
Surgery to remove the cancer can be used alone or in combination with other treatments.

Procedures used for esophageal cancer may include:

- **Removing the cancer from the inside of the esophagus.** Endoscopic resection is a procedure to remove the cancer

and some of the healthy tissue around it. The procedure is done through a long, flexible tube, called an endoscope. The tube goes down the throat and into the esophagus. Special tools are passed through the endoscope to remove the cancer. This procedure might be an option if the cancer is very small and hasn't spread.

- **Removing the cancer and part of the esophagus.** Esophagectomy is surgery to remove part of the esophagus. During esophagectomy, the surgeon removes the part of the esophagus that contains the cancer and some nearby lymph nodes. The surgeon also may remove some of the upper part of the stomach. When the surgery involves removing some of the esophagus and some of the stomach it's called an esophagogastrectomy. The remaining esophagus is reconnected to the stomach. Usually this is done by pulling the stomach up to meet the remaining esophagus.



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Chemotherapy:

Chemotherapy treats cancer with strong medicines. Chemotherapy medicines are typically used before or after surgery in people with esophageal cancer. Chemotherapy can be combined with radiation therapy.

Radiation therapy

Radiation therapy treats cancer with powerful energy beams. The energy can come from X-rays, protons or other sources. For esophageal cancer, radiation therapy is most often done with a procedure called external beam radiation. During this treatment, you lie on a table while a machine moves around you. The machine directs radiation to precise points on your body. Radiation also can be placed inside your body near the cancer. This kind of radiation therapy, called brachytherapy, is less common.

Combined chemotherapy and radiation

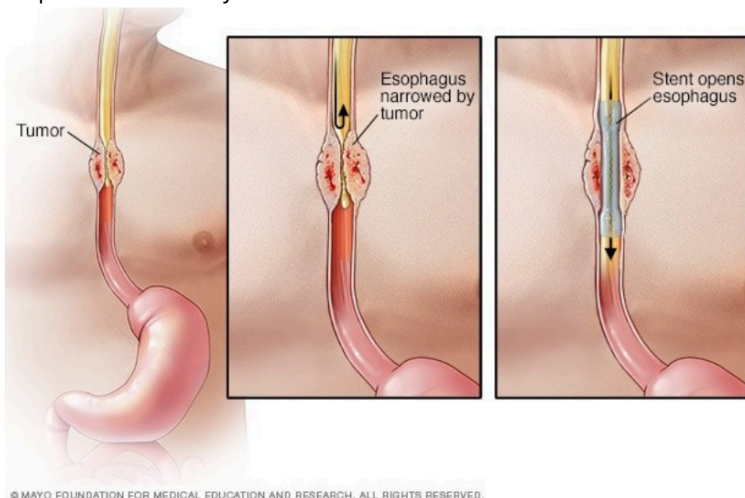
Combining chemotherapy and radiation therapy may enhance the effectiveness of each treatment. Combined chemotherapy and radiation may be the only treatment you receive, or combined therapy can be used before surgery. But combining chemotherapy and radiation treatments increases the likelihood and severity of side effects.

Targeted drug therapy

Targeted therapy for cancer is a treatment that uses medicines that attack specific chemicals in cancer cells. By blocking these chemicals, targeted treatments can cause cancer cells to die.

Immunotherapy

Immunotherapy for cancer is a treatment with medicine that helps the body's immune system kill cancer cells. The immune system fights off diseases by attacking germs and other cells that shouldn't be in the body. Cancer cells survive by hiding from the immune system. Immunotherapy helps the immune system cells find and kill the cancer cells.





UNIT 3 OBT356

Information Technology (Anna University)



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UNIT III

CARDIOVASCULAR DISEASES

Coronary atherosclerosis – Coronary artery disease; Causes -Fat and lipids, Alcohol abuse -- Diagnosis - Electrocardiograph, echocardiograph, Treatment, Exercise and Cardiac rehabilitation

CARDIOVASCULAR DISEASES

Cardiovascular disease (CVD) is a general term for conditions affecting the heart or blood vessels.

It's usually associated with a build-up of fatty deposits inside the arteries (atherosclerosis) and an increased risk of blood clots.

It can also be associated with damage to arteries in organs such as the brain, heart, kidneys and eyes.

CVD is one of the main causes of death and disability but it can often largely be prevented by leading a healthy lifestyle.

Types of CVD

There are many different types of CVD. 4 of the main types are.

1. Coronary heart disease

Coronary heart disease occurs when the flow of oxygen-rich blood to the heart muscle is blocked or reduced.

This puts an increased strain on the heart, and can lead to:

- angina – chest pain caused by restricted blood flow to the heart muscle
- heart attacks – where the blood flow to the heart muscle is suddenly blocked
- heart failure – where the heart is unable to pump blood around the body properly

2. Strokes and TIAs

A stroke is where the blood supply to part of the brain is cut off, which can cause brain damage and possibly death.

A transient ischaemic attack (also called a TIA or "mini-stroke") is similar, but the blood flow to the brain is only temporarily disrupted.

The main symptoms of a stroke or TIA can be remembered with the word FAST, which stands for:

- **Face** – the face may have drooped on one side, the person may be unable to smile, or their mouth or eye may have dropped.
- **Arms** – the person may not be able to lift both arms and keep them there because of arm weakness or numbness in one arm.
- **Speech** – their speech may be slurred or garbled, they may not be able to talk at all or they may not be able to understand what you are saying to them.
- **Time** – it's time to dial 999 immediately if you see any of these signs or symptoms.

3. Peripheral arterial disease

Peripheral arterial disease occurs when there's a blockage in the arteries to the limbs, usually the legs.

This can cause:

- dull or cramping leg pain, which is worse when walking and gets better with rest
- hair loss on the legs and feet
- numbness or weakness in the legs
- persistent ulcers (open sores) on the feet and legs

4.Aortic disease

Aortic diseases are a group of conditions affecting the aorta. This is the largest blood vessel in the body, which carries blood from the heart to the rest of the body.

One of most common aortic diseases is an aortic aneurysm, where the aorta becomes weakened and bulges outwards.

This doesn't usually have any symptoms, but there's a chance it could burst and cause life-threatening bleeding.

Causes of CVD

The exact cause of CVD isn't clear, but there are lots of things that can increase your risk of getting it. These are called "risk factors".

The more risk factors you have, the greater your chances of developing CVD.

If you're over 40, you'll be invited by your GP for an [NHS Health Check](#) every 5 years.

Part of this check involves assessing your individual CVD risk and advising you how to reduce it if necessary.

1.High blood pressure

High blood pressure (hypertension) is one of the most important risk factors for CVD. If your blood pressure is too high, it can damage your blood vessels.

2.Smoking

Smoking and other tobacco use is also a significant risk factor for CVD. The harmful substances in tobacco can damage and narrow your blood vessels.

3.High cholesterol

Cholesterol is a fatty substance found in the blood. If you have high cholesterol, it can cause your blood vessels to narrow and increase your risk of developing a blood clot.

4.Diabetes

Diabetes is a lifelong condition that causes your blood sugar level to become too high.

High blood sugar levels can damage the blood vessels, making them more likely to become narrowed. Many people with type 2 diabetes are also overweight or obese, which is also a risk factor for CVD.

5.Inactivity

If you don't exercise regularly, it's more likely that you'll have high blood pressure, high cholesterol levels and be overweight. All of these are risk factors for CVD.

Exercising regularly will help keep your heart healthy. When combined with a healthy diet, exercise can also help you maintain a healthy weight.

6.Being overweight or obese

Being overweight or obese increases your risk of developing diabetes and high blood pressure, both of which are risk factors for CVD.

You're at an increased risk of CVD if:

- your body mass index (BMI) is 25 or above – use the BMI healthy weight calculator to work out your BMI
- you're a man with a waist measurement of 94cm (about 37 inches) or more, or a woman with a waist measurement of 80cm (about 31.5 inches) or more

7.Family history of CVD

If you have a family history of CVD, your risk of developing it is also increased.

You're considered to have a family history of CVD if either:

- your father or brother were diagnosed with CVD before they were 55
- your mother or sister were diagnosed with CVD before they were 65

Tell your doctor or nurse if you have a family history of CVD. They may suggest checking your blood pressure and cholesterol level.

8.Ethnic background

In the UK people of south Asian and Black African or African Caribbean background have an increased risk of getting CVD.

This is because people from these backgrounds are more likely to have other risk factors for CVD, such as high blood pressure or type 2 diabetes.

9.Other risk factors

Other factors that affect your risk of developing CVD include:

- age – CVD is most common in people over 50 and your risk of developing it increases as you get older
- gender – men are more likely to develop CVD at an earlier age than women
- diet – an unhealthy diet can lead to high cholesterol and high blood pressure
- alcohol – excessive [alcohol](#) consumption can also increase your cholesterol and blood pressure levels, and contribute to weight gain

Preventing CVD

A healthy lifestyle can lower your risk of CVD. If you already have CVD, staying as healthy as possible can reduce the chances of it getting worse.

1.Stop smoking

If you smoke, you should try to give up as soon as possible.

2.Have a balanced diet

A healthy, balanced diet is recommended for a healthy heart.

A balanced diet includes:

- low levels of saturated fat – try to include healthier sources of fat, such as oily fish, nuts and seeds, and olive oil, and avoid unhealthy fats such as fatty cuts of meat, lard, cream, cakes and biscuits
- low levels of salt – aim for less than 6g (0.2oz or 1 teaspoon) a day
- low levels of sugar
- plenty of fibre and wholegrain foods
- plenty of fruit and vegetables

3.Exercise regularly

Adults are advised to do at least 150 minutes of moderate activity a week, such as cycling or brisk walking.

If you find it difficult to do this, start at a level you feel comfortable with and gradually increase the duration and intensity of your activity as your fitness improves.

Visit your GP for a health check if you haven't exercised before or you're returning to exercise after a long break.

4.Maintain a healthy weight

If you're overweight or obese, a combination of regular exercise and a healthy diet can help you lose weight.

If you're struggling to lose weight, your GP or practice nurse can help you come up with a weight loss plan and recommend services in your area.

5.Cut down on alcohol

If you drink alcohol, try not to exceed the recommended limit of 14 alcohol units a week for men and women.

If you do drink this much, you should aim to spread your drinking over 3 days or more.

A unit of alcohol is roughly equivalent to half a pint of normal-strength lager or a single measure (25ml) of spirits. A small glass of wine (125ml) is about 1.5 units.

Your GP can give you help and advice if you're finding it difficult to cut down your drinking.

6.Medicine

If you have a particularly high risk of developing CVD due to high blood cholesterol, your GP may recommend taking medicines called statins to reduce your risk.

Link to study CVD

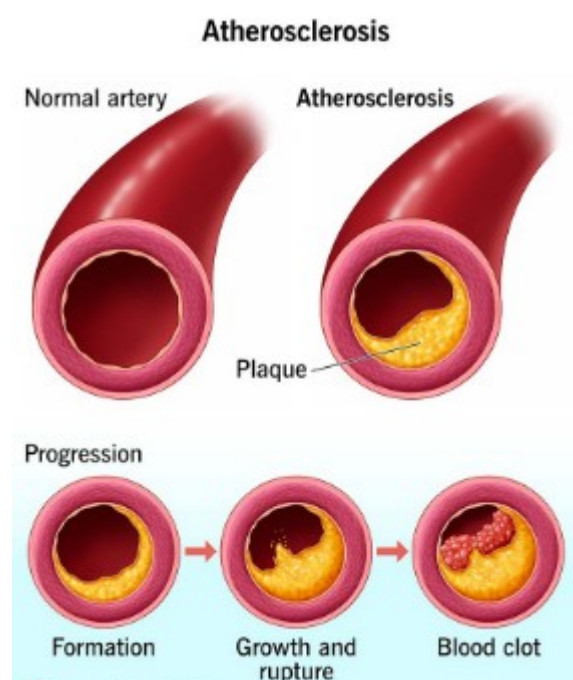
<https://www.youtube.com/watch?v=qJq5hA4pnOk>

CORONARY ATHEROSCLEROSIS - CORONARY ARTERY DISEASE

Atherosclerosis -- sometimes called hardening of the arteries -- can slowly narrow the arteries which supply oxygen-rich blood to the heart and to your body.

When atherosclerosis affects arteries that carry blood to the heart muscle, it's called coronary artery disease, or CAD. That's the No. 1 killer. Most of those deaths are from heart attacks caused by blood clots.

Atherosclerosis can create life-threatening blockages -- without you ever feeling a thing. Since we're all at risk for coronary artery disease, it's worth learning more about atherosclerosis.



Causes of CAD

Clogged arteries lead to heart attacks.

- Cigarette smoking
- High bad cholesterol
- Low good cholesterol
- Obesity
- Lack of physical activity
- High blood pressure
- Diabetes
- Low fruit and vegetable consumption
- Poor socioeconomic status
- Tobacco
- No proper sleep
- Use of hormonal birth control
- Chronic kidney disease

Symptoms:



- You may have no symptoms of coronary artery disease for a long time. Plaque buildup takes many years, even decades. But as your arteries narrow, you may notice mild symptoms. These symptoms mean your heart is pumping harder to deliver oxygen-rich blood to your body
- Some people feel short of breath during light physical activity
- Stable angina/ temporary - chest pain or discomfort that comes and goes in a predictable pattern
- Heart attack

Complications

- The main complication of coronary artery disease is a heart attack. This is a medical emergency that can be fatal. Your heart muscle starts to die because it's not receiving enough blood. You need prompt medical attention to restore blood flow to your heart and save your life
- Over the years, CAD can also weaken your heart and lead to complications, including
- Abnormal heart rhythms
- Cardiac arrest
- Cardiogenic shock
- Heart failure

Diagnosis and Tests

Tests to help diagnose or watch coronary artery disease include:

Blood tests. Blood tests can check blood sugar and cholesterol levels. A high-sensitivity C-reactive protein (CRP) test checks for a protein linked to inflammation of the arteries.

- **Electrocardiogram (ECG or EKG).** This quick test checks the electrical activity of the heart. It shows how the heart is beating. Sticky patches called electrodes attach to the chest and sometimes the arms and legs. Wires connect the electrodes to a computer, which prints or displays the test results. The ECG signal patterns can show if you had or are having a heart attack.
- **Echocardiogram.** This test uses sound waves to show blood flow through the heart. Parts of the heart that move weakly may be caused by a lack of oxygen or a heart attack. This may be a sign of coronary artery disease or other conditions.
- **Exercise stress test.** If your symptoms usually occur during exercise, your healthcare professional may recommend this test. You walk on a treadmill or ride a stationary bike while your heart is checked. Because exercise makes the heart pump harder and faster than it does during most daily activities, an exercise stress test can show heart problems that might otherwise be missed. If you can't exercise, you may be given a medicine that affects the heart like exercise does. Sometimes an echocardiogram is done during an exercise stress test.
- **Nuclear stress test.** This test shows how blood moves to the heart at rest and during activity. It uses a small amount of radioactive material, called a tracer or radiotracer. The substance is given by IV. An imaging machine takes pictures of how the tracer moves through the heart arteries. This helps find areas of poor blood flow or heart damage.
- **Heart CT scan.** A CT scan of the heart can show calcium deposits and blockages in the heart arteries. Calcium deposits can narrow the arteries. Sometimes dye is given by IV during this test. The dye helps create detailed pictures of the heart arteries. If dye is used, the test is called a CT coronary angiogram.
- **Cardiac catheterization and angiogram.** This test can see blockages in the heart arteries. A doctor places a long, thin flexible tube called a catheter in a blood vessel, usually in the groin or wrist. It's guided to the heart. Dye flows through the catheter to arteries in the heart. The dye helps the arteries show up more clearly on X-ray images and video. Heart treatments may be done during this test.

Treatment

Treatment for coronary artery disease may include:

- Lifestyle changes such as not smoking, eating healthy and exercising more.
- Heart procedure or heart surgery.
- **Medications**

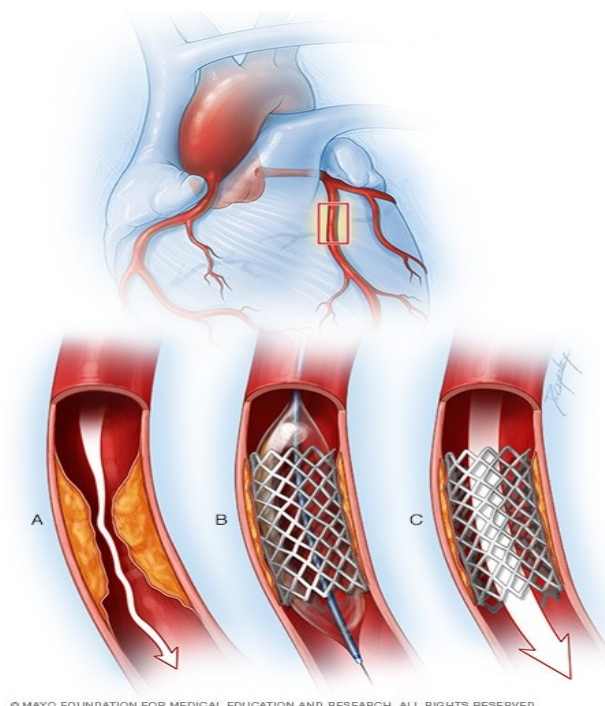
Many medicines are available to treat coronary artery disease,

- **Cholesterol medicine.** Your healthcare professional might recommend this type of medicine to lower "bad" LDL cholesterol and reduce plaque buildup in the arteries. Such medicines include statins, niacin, fibrates and bile acid sequestrants.
- **Aspirin.** Aspirin helps thin the blood and prevent blood clots. Daily low-dose aspirin therapy may be recommended for the primary prevention of heart attack or stroke in some people.

Daily use of aspirin can have serious side effects, including bleeding in the stomach and intestines. Don't start taking a daily aspirin without talking to your healthcare team.

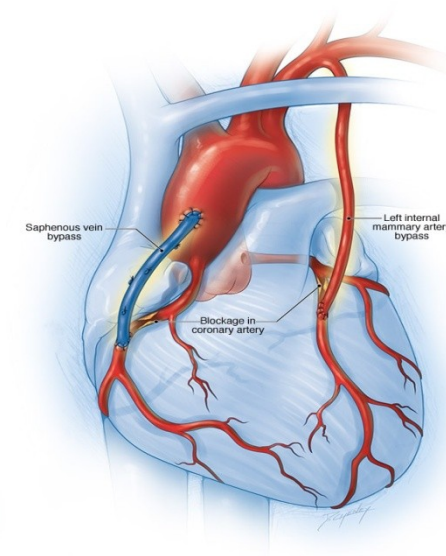
- **Beta blockers.** These medicines slow the heartbeat and lower blood pressure. If you've had a heart attack, beta blockers may reduce the risk of future heart attacks.
- **Calcium channel blockers.** One of these medicines may be suggested if you can't take beta blockers or beta blockers don't work for you. Calcium channel blockers can help reduce chest pain.
- **Angiotensin-converting enzyme (ACE) inhibitors and angiotensin 2 receptor blockers (ARBs).** These medicines lower blood pressure. They may help keep coronary artery disease from getting worse.
- **Nitroglycerin.** This medicine widens the heart arteries. It can help control or reduce chest pain. Nitroglycerin is available as a pill, spray or patch.
- **Ranolazine.** This medicine may help people with long-term chest pain. It may be prescribed with or instead of a beta blocker.

Surgeries or other procedures



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Coronary artery stent



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Coronary artery bypass surgery

Surgery may be done to fix a blocked artery and improve blood flow. Surgeries or procedures for coronary artery disease may include:

- **Coronary angioplasty and stent placement.** This treatment opens clogged blood vessels in the heart. A tiny balloon on a thin tube, called a catheter, is used to widen a clogged artery and improve blood flow. A small wire mesh tube called a stent may be placed to keep the artery open. Most stents are coated with medicine that helps keep the artery open. This treatment also is called percutaneous coronary intervention.
- **Coronary artery bypass graft (CABG) surgery.** This is a type of open-heart surgery. During CABG, a surgeon takes a vein or artery from somewhere else in the body. The surgeon uses the blood vessel to create a new path for blood to go around a blocked or narrowed heart artery. The surgery increases blood flow to the heart.

If a patient had coronary artery bypass surgery, your healthcare professional may suggest cardiac rehabilitation. This is a program of education, counseling and exercise training that's designed to help improve the health after heart surgery.

ELECTROCARDIOGRAM

An electrocardiogram (ECG or EKG) is one of the simplest and fastest tests used to evaluate the heart. Electrodes (small, plastic patches that stick to the skin) are placed at certain spots on the chest, arms, and legs. The electrodes are connected to an ECG machine by lead wires. The electrical activity of the heart is then measured, interpreted, and printed out. No electricity is sent into the body.

Natural electrical impulses coordinate contractions of the different parts of the heart to keep blood flowing the way it should. An ECG records these impulses to show how fast the heart is beating, the rhythm of the heart beats (steady or irregular), and the timing of the electrical impulses as they move through the different parts of the heart. Changes in an ECG can be a sign of many heart-related conditions.

Principle:

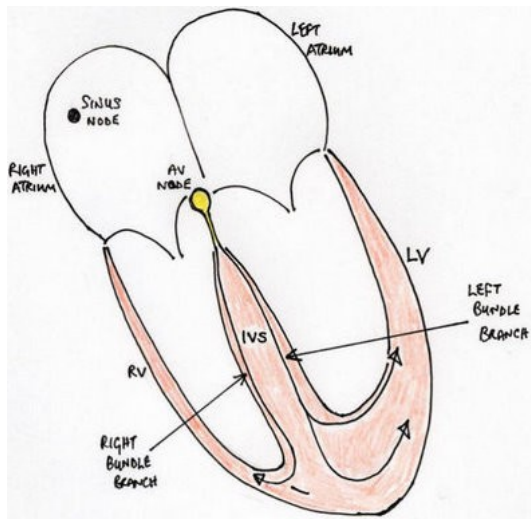
With modern machines, surface ECGs are quick and easy to obtain at the bedside and are based on relatively simple electrophysiological concepts. However junior doctors often find them difficult to interpret.

An ECG is simply a representation of the electrical activity of the heart muscle as it changes with time, usually printed on paper for easier analysis. Like other muscles, cardiac muscle contracts in response to electrical *depolarisation* of the muscle cells. It is the sum of this electrical activity, when amplified and recorded for just a few seconds that we know as an ECG.

The normal cardiac cycle begins with spontaneous depolarisation of the sinus node, an area of specialised tissue situated in the high right atrium (RA). A wave of electrical depolarisation then spreads through the RA and across the inter-atrial septum into the left atrium (LA).

The atria are separated from the ventricles by an electrically inert fibrous ring, so that in the normal heart the only route of transmission of electrical depolarisation from atria to ventricles is through the atrioventricular (AV) node. The AV node delays the electrical signal for a short time, and then the wave of depolarisation spreads down the interventricular septum (IVS), via the bundle of His and the right and left bundle branches, into the right (RV) and left (LV) ventricles. Hence with normal conduction the two ventricles contract simultaneously, which is important in maximising cardiac efficiency.

After complete depolarisation of the heart, the myocardium must then *repolarise*, before it can be ready to depolarise again for the next cardiac cycle.



Basic electrophysiology of the heart

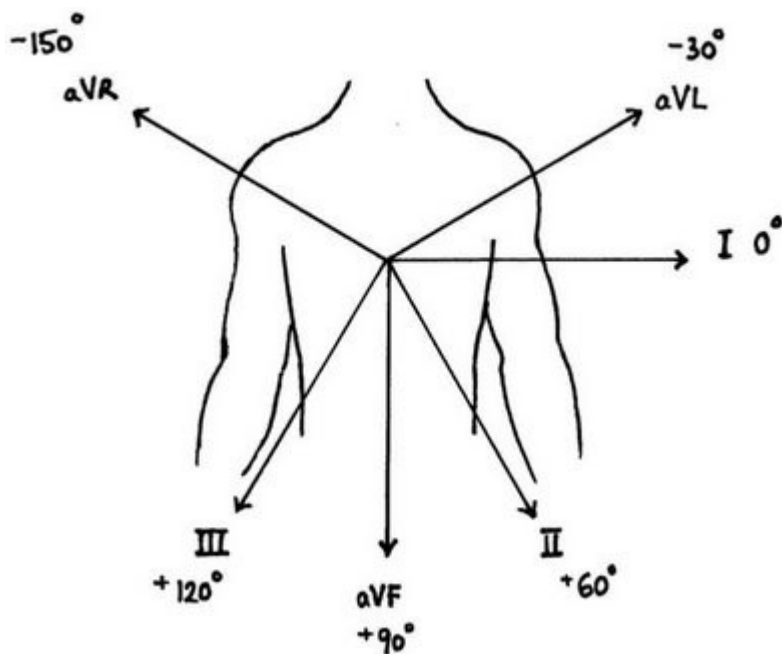
Measurement:

The ECG is measured by placing a series of electrodes on the patient's skin – so it is known as the 'surface' ECG.

The wave of electrical depolarisation spreads from the atria down through the IVS to the ventricles. So the direction of this depolarisation is usually from the superior to the inferior aspect of the heart. The direction of the wave of depolarisation is normally towards the left due to the leftward orientation of the heart in the chest and the greater muscle mass of the left ventricle than the right. This overall direction of travel of the electrical depolarisation through the heart is known as the *electrical axis*.

A fundamental principle of ECG recording is that when the wave of depolarisation travels toward a recording lead this results in a positive or upward deflection. When it travels away from a recording lead this results in a negative or downward deflection.

The electrical axis is normally downward and to the left but we can estimate it more accurately in individual patients if we understand from which 'direction' each recording lead measures the ECG.



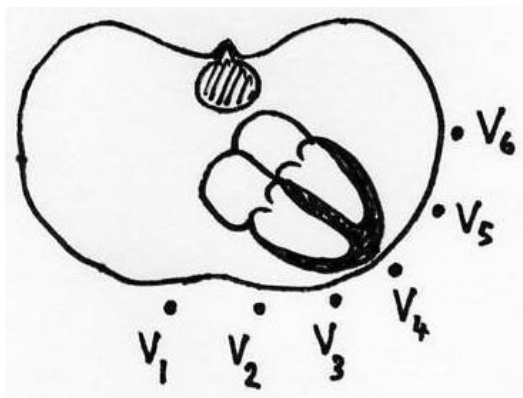
Orientation of the limb leads showing the direction from which each lead 'looks' at the heart

By convention, we record the standard surface ECG using 12 different recording lead 'directions,' though rather confusingly only 10 recording electrodes on the skin are required to achieve this. Six of these are recorded from the chest overlying the heart – *the chest or precordial leads*. Four are recorded from the limbs – *the limb leads*. It is essential that each of the 10 recording electrodes is placed in its correct position, otherwise the appearance of the ECG will be changed significantly, preventing correct interpretation.

The limb leads record the ECG in the coronal plane, and so can be used to determine the electrical axis (which is usually measured only in the coronal plane). The limb leads are called leads I, II, III, AVR, AVL and AVF. Figure 2 shows the relative directions from which they 'look' at the heart. A horizontal line through the heart and directed to the left (exactly in the direction of lead I) is conventionally labelled as the reference point of 0 degrees (0°). The directions from which other leads 'look' at the heart are described in terms of the angle in degrees from this baseline.

The electrical axis of depolarisation is also expressed in degrees and is normally in the range from -30° to +90°. A detailed explanation of how to determine the axis is beyond the scope of this article but the principles mentioned here should help readers to understand the concepts involved.

The chest leads record the ECG in the transverse or horizontal plane, and are called V1, V2, V3, V4, V5 and V6



Transverse section of the chest showing the orientation of the six chest leads in relation to the heart

Voltage and timing intervals

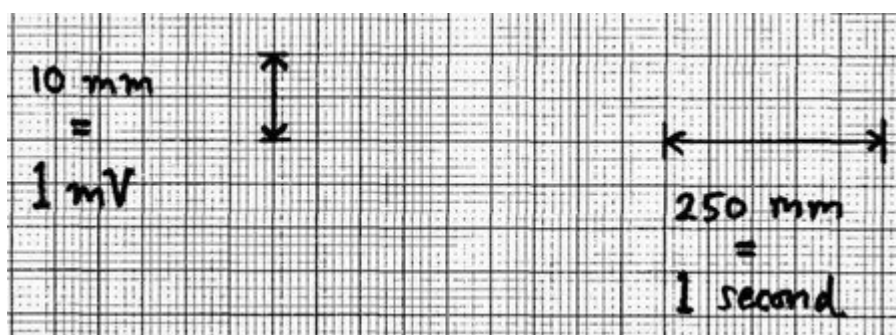
It is conventional to record the ECG using standard measures for amplitude of the electrical signal and for the speed at which the paper moves during the recording. This allows:

- Easy appreciation of heart rates and cardiac intervals and
- Meaningful comparison to be made between ECGs recorded on different occasions or by different ECG machines.

The amplitude, or voltage, of the recorded electrical signal is expressed on an ECG in the vertical dimension and is measured in millivolts (mV). On standard ECG paper 1mV is represented by a deflection of 10 mm. An increase in the amount of muscle mass, such as with left ventricular hypertrophy (LVH), usually results in a larger electrical depolarisation signal, and so a larger amplitude of vertical deflection on the ECG.

An essential feature of the ECG is that the electrical activity of the heart is shown as it varies with time. In other words we can think of the ECG as a graph, plotting electrical activity on the vertical axis against time on the horizontal axis. Standard ECG paper moves at 25 mm per second during real-time recording. This means that when looking at the printed ECG a distance of 25 mm along the horizontal axis represents 1 second in time.

ECG paper is marked with a grid of small and large squares. Each small square represents 40 milliseconds (ms) in time along the horizontal axis and each larger square contains 5 small squares, thus representing 200 ms. Standard paper speeds and square markings allow easy measurement of cardiac timing intervals. This enables calculation of heart rates and identification of abnormal electrical conduction within the heart



Sample of standard ECG paper showing the scale of voltage, measured on the vertical axis, against time on the horizontal axis

The normal ECG

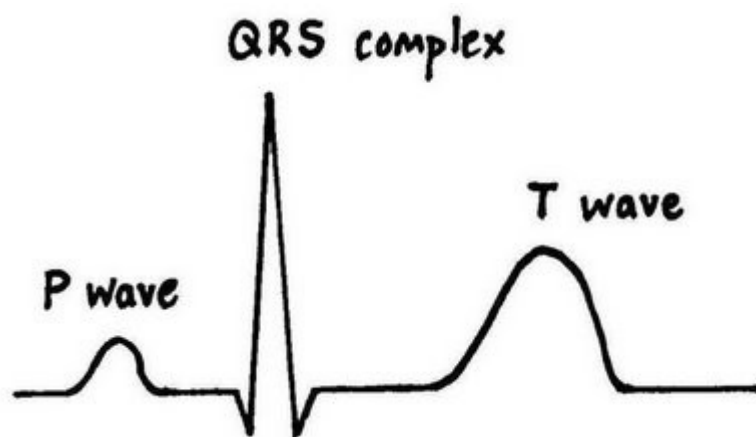
It will be clear from above that the first structure to be depolarised during normal sinus rhythm is the right atrium, closely followed by the left atrium. So the first electrical signal on a normal ECG originates from the atria and is known as the **P wave**. Although there is usually only one P wave in most leads of an ECG, the P wave is in fact the sum of the electrical signals from the two atria, which are usually superimposed.

There is then a short, physiological delay as the atrioventricular (AV) node slows the electrical depolarisation before it proceeds to the ventricles. This delay is responsible for the PR interval, a short period where no electrical activity is seen on the ECG, represented by a straight horizontal or 'isoelectric' line.

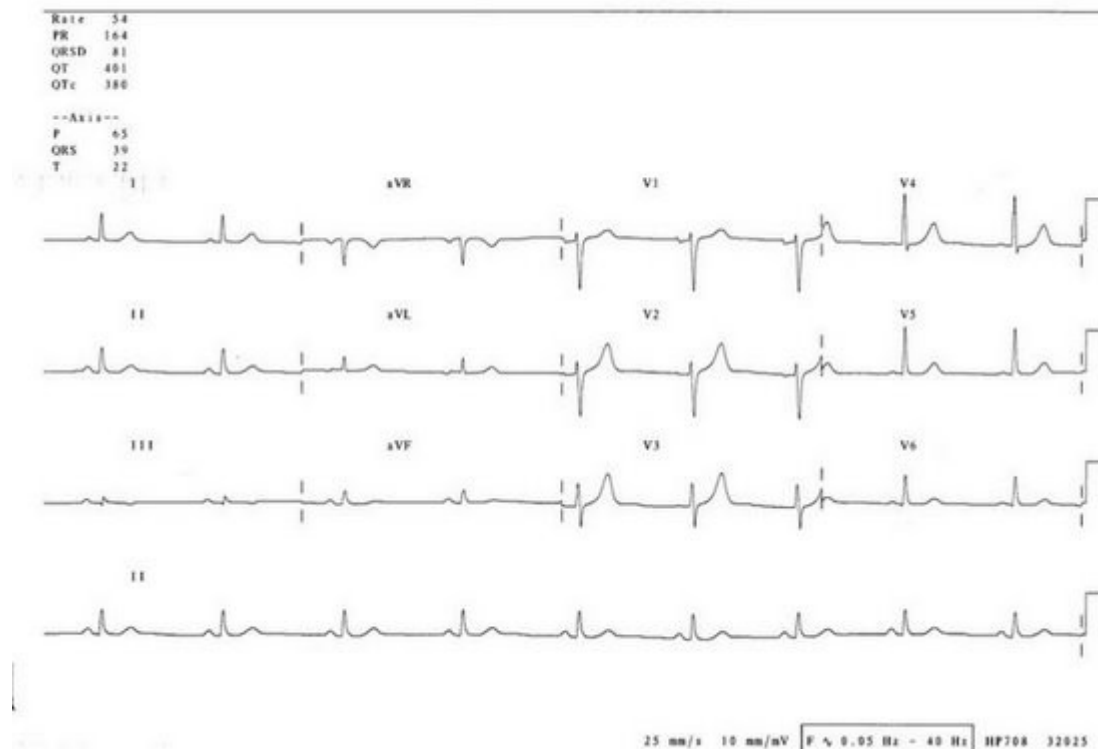
Depolarisation of the ventricles results in usually the largest part of the ECG signal (because of the greater muscle mass in the ventricles) and this is known as the **QRS complex**.

- The Q wave is the first initial downward or 'negative' deflection
- The R wave is then the next upward deflection (provided it crosses the isoelectric line and becomes 'positive')
- The S wave is then the next deflection downwards, provided it crosses the isoelectric line to become briefly negative before returning to the isoelectric baseline.

In the case of the ventricles, there is also an electrical signal reflecting repolarisation of the myocardium. This is shown as the **ST segment** and the **T wave**. The ST segment is normally isoelectric, and the T wave in most leads is an upright deflection of variable amplitude and duration



The major waves of a single normal ECG pattern



Example of a normal 12 lead ECG; notice the downward deflection of all signals recorded from lead aVR. This is normal, as the electrical axis is directly away from that lead

Normal intervals

The recording of an ECG on standard paper allows the time taken for the various phases of electrical depolarisation to be measured, usually in milliseconds. There is a recognised normal range for such 'intervals':

- **PR interval** (measured from the beginning of the P wave to the first deflection of the QRS complex). Normal range 120 – 200 ms (3 – 5 small squares on ECG paper).
- **QRS duration** (measured from first deflection of QRS complex to end of QRS complex at isoelectric line). Normal range up to 120 ms (3 small squares on ECG paper).
- **QT interval** (measured from first deflection of QRS complex to end of T wave at isoelectric line). Normal range up to 440 ms (though varies with heart rate and may be slightly longer in females)

Heart rate estimation from the ECG

Standard ECG paper allows an approximate estimation of the heart rate (HR) from an ECG recording. Each second of time is represented by 250 mm (5 large squares) along the horizontal axis. So if the number of large squares between each QRS complex is:

- 5 - the HR is 60 beats per minute.
- 3 - the HR is 100 per minute.
- 2 - the HR is 150 per minute.

ECHOCARDIOGRAM

An echocardiogram is an ultrasound image of the heart. Echocardiograms can help doctors diagnose a range of heart problems, such as heart attacks, blood clots, heart valve disease, and more.

Doctors use echocardiograms to help them diagnose heart problems, such as damaged cardiac tissue, chamber enlargement, stiffening of the heart muscle, blood clots in the heart, fluid around the heart, and damaged or poorly functioning heart valves. It is a noninvasive medical procedure that produces no radiation and does not typically cause side effects.

During an echocardiogram, a doctor can see:

- the size and thickness of the chambers
- how the valves of the heart are functioning
- the direction of blood flow through the heart
- any blood clots in the heart
- areas of damaged or weak cardiac muscle tissue
- problems affecting the pericardium, which is the fluid-filled sac around the heart
- causes of a stroke

Doctors also use echocardiography when they want to examine a person's general heart health, especially after a heart attack or stroke.

Echocardiograms are noninvasive and relatively quick procedures that require minimal preparation.

Below, we discuss what to expect before, during, and after an echocardiogram.

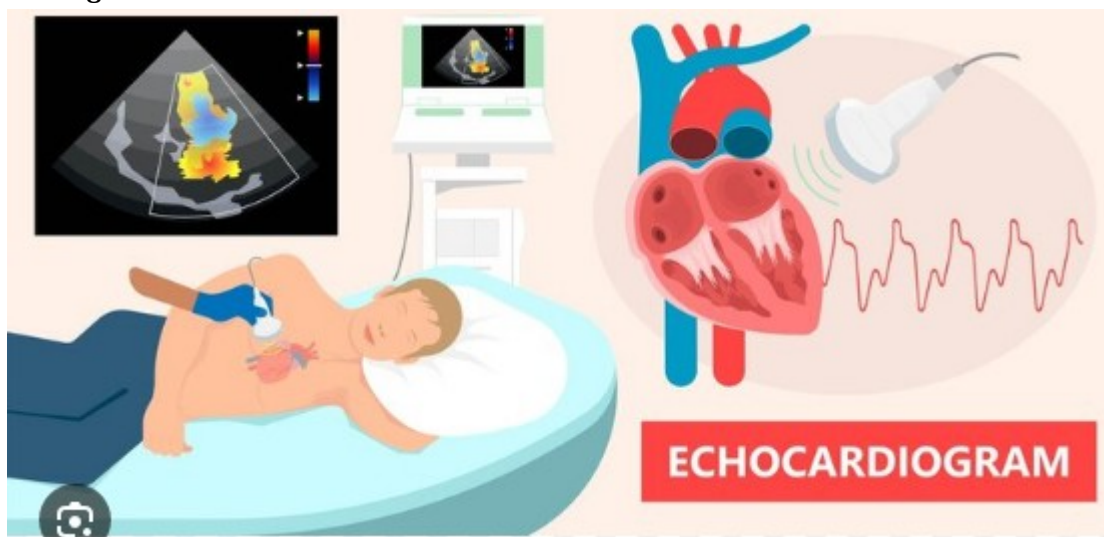
Method:

Preparation

In cases where a healthcare professional takes the echocardiogram from the outside of the body, the person will not need to prepare.

For people who get a transesophageal echocardiogram, a doctor will recommend [Trusted Source](#) avoiding eating or drinking anything for at least 6 hours before the exam. People can resume eating and drinking about 1–2 hours after the echocardiogram once the local anesthetic has worn off.

During the test



A sonographer will perform the transthoracic (external) echocardiogram. Sonographers are healthcare professionals who specialize in using ultrasound devices to produce images and videos for diagnostic purposes.

During the test, the person receiving the echocardiogram will remove their clothes from the waist up. They can wear a hospital gown if they wish to cover themselves during the exam.

The sonographer will then instruct the person to lie on a table, on either their back or their left side. They may inject a saline solution or dye into the person's veins, which makes the heart appear more defined on the echocardiogram.

The exact procedure depends on the type of echocardiogram.

For instance:

Transthoracic echocardiogram

If a doctor ordered a transthoracic echocardiogram, the sonographer will apply a gel to the chest. The sonographer will then move the transducer around the chest to get different images of the heart.

During the exam, the sonographer may ask someone to change positions or take or hold a deep breath. They might press the transducer into the chest to get a better picture of the heart.

Transesophageal echocardiogram

A doctor might order a transesophageal echocardiogram if they want more detailed or clearer images of the heart than those that a transthoracic echocardiogram can produce. During a transesophageal echocardiogram, the person may receive a mild sedative to help relax the muscles in their throat, and a local anesthetic to numb the gag reflex.

Once the sedative and local anesthetic take effect, a doctor will guide a small transducer on the end of a long tube down the throat and esophagus until it reaches the back of the heart.

The sonographer will record images of the heart as the doctor moves the transducer around the esophagus. The person should not feel the transducer or the tube in their esophagus after initially swallowing the probe.

After the test

Most people can return to their regular activities after having a transthoracic echocardiogram.

People who have a transesophageal echocardiogram may need to stay at the hospital or healthcare clinic for a few hours after the exam. They may have a sore throat initially, but it should improve within a few hours to a day.

Individuals who received a sedative before the exam should not drive for several hours after the echocardiogram.

What does it diagnose?

Doctors can use echocardiograms to see the size, structure, and activity of various parts of the heart.

They do this to diagnose heart problems, determine the need for more tests, decide on their next actions, and monitor changes and improvements.

More specifically, doctors may use this procedure to check for signs or symptoms that may be indicative of:

- **Heart attack:** the test can check for impaired blood supply in heart muscle tissue, wall abnormalities, and blood flow, which can indicate a heart attack.
- **Blood clots (thrombus) or tumors:** A 2021 study found that echo can be an alternative to cardiac magnetic resonance in detecting thrombosis. A 2020 study also saw it as an essential noninvasive tool in checking for cardiac masses such as tumors.
- **Atherosclerosis and coronary artery disease (CAD):** While an echo cannot show blockages in arteries, narrowing and clogged arteries can affect the heart's

pumping ability and wall motion. This is more evident during stress, making a stress echo a good diagnostic test.

- **Aortic aneurysm and aortic dissection:** An echo can screen [Trusted Source](#) for wide, weakened aorta, unruptured aneurysms, and their size, as well as the formation of fibrosis and thrombus in the vessel.
- **Cardiomyopathy:** The test can see the size and function of the heart and correlate it with wall thickness, weak heart muscle, leaky heart valves, heart failure, or high blood pressure.
- **Pulmonary hypertension:** The test can assess [Trusted Source](#) the pressure in the heart, which can indicate the presence of pulmonary hypertension, helping doctors to determine the next diagnostic steps.
- **Congenital heart disease:** The test can identify congenital heart abnormalities in infants and young children, such as septal defects and holes.
- **Heart valve disease:** The test looks for abnormalities in heart blood flow, leakage, narrowing, infection, and blockage in heart valves.
- **Problems with the pericardium:** The test can check the status of the sac surrounding the heart (pericardium) for inflammation (pericarditis) or becoming filled with fluid or blood (pericardial effusion).
- **Heart failure:** It can detect weak or stiff and thickened heart muscle, which can be a sign of heart failure.

Doctors also often use the test to assess the reasons for an abnormal electrical test of the heart, called an electrocardiogram (EKG).

They also use the procedure to monitor how well the heart responds to different heart treatments, such as heart failure, medications, artificial valves, and pacemakers.

A doctor will order an echocardiogram if they suspect that someone has heart problems.

Signs and symptoms that may indicate a heart condition include:

- an irregular heartbeat (arrhythmia)
- shortness of breath
- high or low blood pressure
- leg swelling
- abnormal EKG results
- unusual sounds between heartbeats, known as heart murmurs [Trusted Source](#)

Types of echocardiogram

Doctors can order different types of echocardiograms, all of which use high-frequency sound waves. The common types include those below.

The transthoracic echocardiogram is the most common [Trusted Source](#) type of echocardiogram test.

This test involves placing an ultrasound wand called a transducer on the outside of the chest, near the heart. The device sends sound waves through the chest and into the heart.

Applying a gel to the chest helps the sound waves travel better. These waves bounce off the heart and create images of the heart structures on a screen.

Transesophageal echocardiogram

A transesophageal echocardiogram uses a thinner transducer that attaches to the end of a long tube. The individual will swallow the tube to insert it into the esophagus, the organ that connects the mouth and stomach, which runs behind the heart.

This type of echocardiogram provides more detailed pictures of the heart compared with the traditional transthoracic echocardiogram because it gives a “close up” view of this organ.

Doppler ultrasound

Doctors use doppler ultrasounds to check the flow of blood. They do this by generating sound waves at specific frequencies and determining how the sound waves bounce off and return to the transducer.

Doctors can use color doppler ultrasounds to map the direction and velocity of blood flow in the heart. Blood that flows toward the transducer appears red, while blood that flows away looks blue. It can also determine the degree of blockages.

The results of a doppler ultrasound can reveal problems with valves or holes in the walls of the heart and help doctors assess how the blood is traveling through it.

Learn more about a Doppler ultrasound [here](#).

3D echocardiogram

A 3D echocardiogram creates detailed 3D images of the heart. Doctors can use 3D echocardiograms to:

- assess valve functionality in people who have heart failure
- diagnose heart problems in infants and children
- plan heart valve or structural interventional surgery
- assess the function of the heart in 3D
- image complex structures within the heart

Stress echocardiogram

A doctor can order an echocardiogram as part of a stress test. A stress test [Trusted Source](#) involves physical exercise, such as walking, jogging on a treadmill, or riding a bike.

During the test, the doctor will monitor heart rate, blood pressure, and the heart's electrical activity.

A sonographer will take a transthoracic echocardiogram before and after the exercise.

Doctors use stress tests to diagnose:

- ischemic heart disease
- coronary heart disease
- heart failure
- problems affecting the heart valves

Point-of-care (POC) echocardiogram

A POC echocardiogram is a type of echocardiogram that a doctor can conduct at a person's bedside. These can help to answer specific questions a doctor may have regarding potential differential diagnoses.

There are two types [Trusted Source](#) of POC echo — limited and focused.

A limited echocardiogram helps a doctor identify the cause and effects of the injury that affected the heart. A doctor performs a focused echo to help to narrow down the list of other potential diagnoses or answer a particular question they may have.

A POC echocardiogram can assess how well the left or right ventricles pump the blood with each heartbeat.

Fetal echocardiogram

Doctors can use a fetal echocardiogram to view an unborn baby's heart. This exam usually takes place at about 18–22 weeks [Trusted Source](#) of pregnancy. Echocardiograms do not use radiation, so they are not harmful to the mother or baby.

Interpreting the results

After the exam, the sonographer will send the echocardiographic images to the doctor who ordered the test. The doctor will review the images and look for signs of heart problems, such as:

- damaged heart muscle tissue
- pumping function of the heart
- thick or thin ventricle walls
- abnormal chamber size
- poorly functioning valves
- chamber size
- masses in the heart, such as blood clots or tumors

Drawbacks

Echocardiograms are especially helpful in looking at structural problems of the heart. However, they may not be the best procedure to check the coronary arteries.

Blockages can cause changes in the heart structure. Doctors can usually detect changes in the heart function or weak muscles or thinner heart walls, which cause them to run further tests such as a coronary angiogram.

Echocardiograms also cannot check for conduction disorders or electrical problems of the heart that affect its rhythm, but can assess the effects of these abnormalities on the heart.

Echocardiogram vs. electrocardiogram

People should not confuse an echocardiogram with another diagnostic test called an electrocardiogram or EKG. An EKG measures the electrical impulses or waves that travel through cardiac muscle tissue.

The electrical activity in the heart causes the heart muscle tissues to contract and relax, which creates the rhythmic heartbeat that people can hear through a stethoscope.

A trained technician, nurse, or doctor can take an EKG by placing electrodes on the skin of the chest, arms, or legs. These electrodes record electrical activity and send the information to a computer that converts it into a graph, which a doctor can print out.

DIFFERENCE BETWEEN ECG AND ECHO

	Differences	ECG Test	ECHO Test
1.	Technique	With an ECG/EKG, doctors attach electrodes to various spots on your chest wall to record heart activity and interpret it on the tracing graph.	In an ECHO, doctors apply ultrasound gel to the chest wall and then place the ultrasound probes on the chest to capture images of your heart and valve structure.
2.	Detailing	An EKG shows cardiologists the peaks and dips of the heart rhythm, providing limited comprehensive information that is more often related to impending heart attacks.	An echo provides profoundly accurate information about the overall structure and function.
3.	Timing	The test usually takes only five minutes to perform.	The test can take anywhere from 20 minutes to an hour, depending on the specific information that the doctor needs.
4.	Popularity	An ECG or EKG is more affordable for many patients. Doctors also often request this test for almost every patient with symptoms of heart disease, or patients with a family history of cardiovascular diseases.	Doctors usually don't request an Echo unless your EKG result returns abnormal findings.

CARDIAC REHABILITATION EXERCISES

- Warm-up or stretching exercises
- Cardio exercises
- Strengthening exercises
- Cool-down exercises

Before starting blood pressure and heart rate are checked, warm-up is done for 15 minutes, then start the main exercise end with the cooldown exercise for 10 minutes and again check the blood pressure and heart rate.

Warm-up:

Always warm-up for 15 minutes with light exercise e.g. walking, marching on the spot, or low-level cycling, followed by stretching of the muscles, that reduces the risk of injuries.

Stretching Exercises Include:

1: Upper back stretch

Lock the fingers together with the arms stretched in front. Lower the head forward to look at the floor with feet moving. Hold the stretch for 10-15 seconds.

2: Chest stretch

Place the hands on the lower back. Gently move the elbows towards each other by keeping the back straight and feet moving. Hold the stretch for 10-15 seconds

3: Calf stretch

Press the heel of the back leg into the floor till a gentle stretch is felt in the back of the lower leg. Adjust the position by moving the back foot further back and the hips forward in case the stretch is not felt. Hold the stretch for 10-15 secs for each leg.

4: Hamstring stretch

Stand and lean forward slightly with one leg in front of the other and hands-on the hips. Slightly bend the back leg, straighten the front leg. Lean until a stretch is felt in the back of the leg placed. Hold for 10-15 seconds. Repeat on the other leg.

5: Front of thigh stretch

Put the left hand on the wall for support. Lift the right knee in front and hold the right ankle. Keeping the back straight, push the hip forward until a gentle stretch is felt. Hold the stretch for 10-15 secs and repeat with the other leg.

5: Pulse raising activities

During the last 5 minutes of warm-up, pulse-raising activities at a slightly higher intensity like walking, marching, cycling, etc are done. This helps to increase the heart rate and prepares the person for the main exercise session.

Cardiovascular Exercises:

After a warm-up and stretching exercises, cardio exercises are started. These exercises strengthen the heart muscles and help them to pump harder and faster. These aerobic exercises include:

- Stationery biking
- Elliptical machine exercising
- Jumping rope
- Walking
- Jogging
- Running
- Swimming
- Dancing
- Playing sports like soccer, tennis, basketball, etc.

Strengthening Exercises

Strengthening exercises keep the body in shape and increase the strength for daily activities like opening a jar or carrying groceries.

1: Monitoring the exercise level

One should ensure that the body is not pushed too hard during the rehabilitation program, there are various ways by which this can be checked, so that the exercises are done safely and effectively, under the supervision of a trained physiotherapist.

2: Measuring your heart rate (pulse)

Heart rate is the no, of times the heartbeats per minute. It is an indicator of how hard the patient is working. During exercise the heart rate increases to fulfill the demand of working muscles to supply blood and oxygen.

The heart rate can be palpated between the tendons in the middle of the wrist and the bone on the outside of the up-turned arm, about 1 inch from the base of the thumb.

- Gently place the index and middle fingers over the area, feeling a slight pulse in the heart pushes blood around the body.
- Count the number of beats that are felt for 15 secs.
- Multiply this by 4 and this will give, how many times per minute the heart beats, this gives the heart rate.
- The heart rate should be measured soon after each exercise before it starts to slow down
- There is a recommended heart rate range when the exercise is started. The target heart rate is just a guide. Certain medications, like beta-blockers, affect the way the heart rate responds to exercise. It might not increase much. In case the top end of this range is reached there is no need to worry just reduce the level.

3: Effort score

Another way of measuring how hard the exercise is, measure the level of effort and give it a score of between 1 and 10.

During the stage of recovery moderate, somewhat strong exercise is recommended, if the effort level is strong, then the exercise is too hard and needs to slow down. The exercise level should be checked and kept within comfortable limits.

4: Talk test

This is another simple way of assessing how hard the exercise is. This test can be done by asking the person to talk, if the person is able to speak in complete sentences, then the exercise level is correct. Shortness of breath or gasp indicates that exercise is too hard and needs to slow down.

COOLDOWN:

Cooldown exercises are done for 10 minutes to bring the body back to its resting state, this reduces the risk of dizziness or fainting that can occur due to a sudden drop in blood pressure as a result of suddenly stopping exercise. Stretching during the cool down also helps to reduce muscle soreness that may be caused by the activity. The cool-down exercise should be performed at a slower speed.

CARDIAC REHABILITATION

Cardiac rehabilitation is done to improve health in those with a heart condition or a history of heart surgery. The goals of cardiac rehabilitation are to:

- Get stronger
- Reduce the risk of future heart problems
- Prevent the heart condition from worsening
- Improve quality of life

Cardiac rehabilitation is an option for people with many forms of heart disease. Your provider may recommend cardiac rehab if your medical history includes:

- Angioplasty and stenting
- Cardiomyopathy
- Certain congenital heart diseases
- Chest pain (stable angina)
- Coronary artery bypass surgery
- Coronary artery disease
- Heart attack
- Heart failure
- Heart or lung transplant
- Heart valve repair or replacement
- Peripheral artery disease with pain in the arms or legs during activity (claudication)



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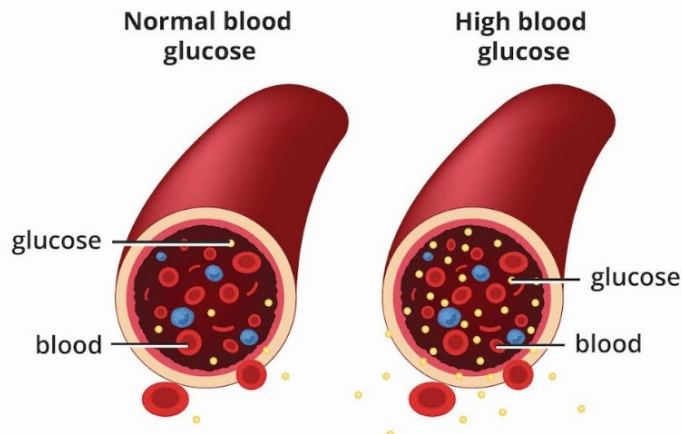
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DIABETES AND OBESITY

Types of Diabetes mellitus; Blood glucose regulation; Complications of diabetes – Paediatric and adolescent obesity – Weight control and BMI

**DIABETES:**

Diabetes is a condition that happens when your blood sugar (glucose) is too high. It develops when your pancreas doesn't make enough insulin or any at all, or when your body isn't responding to the effects of insulin properly. Diabetes affects people of all ages. Most forms of diabetes are chronic (lifelong), and all forms are manageable with medications and/or lifestyle changes

Glucose (sugar) mainly comes from carbohydrates in your food and drinks. It's your body's go-to source of energy. Your blood carries glucose to all your body's cells to use for energy..

When glucose is in your bloodstream, it needs help — a “key” — to reach its final destination. This key is insulin (a hormone). If your pancreas isn't making enough insulin or your body isn't using it properly, glucose builds up in your bloodstream, causing high blood sugar (hyperglycemia).

Over time, having consistently high blood glucose can cause health problems, such as heart disease, nerve damage and eye issues

The technical name for diabetes is diabetes mellitus. Another condition shares the term “diabetes” — diabetes insipidus — but they're distinct. They share the name “diabetes” because they both cause increased thirst and frequent urination. Diabetes insipidus is much rarer than diabetes mellitus.

Types:

There are several types of diabetes. The most common forms include

Type 2 diabetes: With this type, your body doesn't make enough insulin and/or your body's cells don't respond normally to the insulin (insulin resistance). This is the most common type of diabetes. It mainly affects adults, but children can have it as well

Prediabetes: This type is the stage before Type 2 diabetes. Your blood glucose levels are higher than normal but not high enough to be officially diagnosed with Type 2 diabetes

Type 1 diabetes: This type is an autoimmune disease in which your immune system attacks and destroys insulin-producing cells in your pancreas for unknown reasons. Up to 10% of people who have diabetes have Type 1. It's usually diagnosed in children and young adults, but it can develop at any age

Gestational diabetes: This type develops in some people during pregnancy. Gestational diabetes usually goes away after pregnancy. However, if you have gestational diabetes, you're at a higher risk of developing Type 2 diabetes later in life

Type 3c diabetes: This form of diabetes happens when your pancreas experiences damage (other than autoimmune damage), which affects its ability to produce insulin. Pancreatitis, pancreatic cancer, cystic fibrosis and hemochromatosis can all lead to pancreas damage that causes diabetes. Having your pancreas removed (pancreatectomy) also results in Type 3c

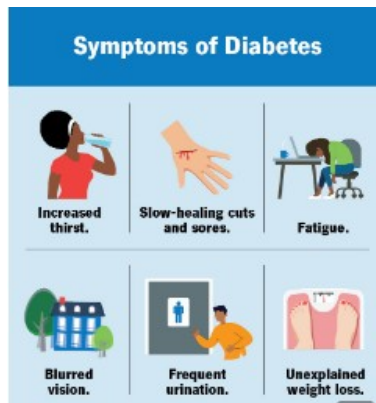
Latent Autoimmune diabetes in Adults[LADA]: Like Type 1 diabetes, LADA also results from an autoimmune reaction, but it develops much more slowly than Type 1. People diagnosed with LADA are usually over the age of 30.

Maturity Onset diabetes of Young[MODY]: MODY, also called monogenic diabetes, happens due to an inherited genetic mutation that affects how your body makes and uses insulin. There are currently over 10 different types of MODY. It affects up to 5% of people with diabetes and commonly runs in families

Neonatal Diabetes: This is a rare form of diabetes that occurs within the first six months of life. It's also a form of monogenic diabetes. About 50% of babies with neonatal diabetes have the lifelong form called permanent neonatal diabetes mellitus. For the other half, the condition disappears within a few months from onset, but it can come back later in life. This is called transient neonatal diabetes mellitus

Brittle diabetes: Brittle diabetes is a form of Type 1 diabetes that's marked by frequent and severe episodes of high and low blood sugar levels. This instability often leads to hospitalization. In rare cases, a pancreas transplant may be necessary to permanently treat brittle diabetes

Symptoms::



- Increased thirst (polydipsia) and dry mouth
- Frequent urination
- Fatigue
- Blurred vision
- Unexplained weight loss
- Numbness or tingling in your hands or feet
- Slow-healing sores or cuts
- Frequent skin and/or vaginal yeast infections

CAUSES:

Insulin Resistance: Type 2 diabetes mainly results from insulin resistance. Insulin resistance happens when cells in your muscles, fat and liver don't respond as they should to insulin. Several factors and conditions contribute to varying degrees of insulin resistance, including obesity, lack of physical activity, diet, hormonal imbalances, genetics and certain medications

Autoimmune Disease: Type 1 diabetes and LADA happen when your immune system attacks the insulin-producing cells in your pancreas

Hormonal Imbalance: During pregnancy, the placenta releases hormones that cause insulin resistance. You may develop gestational diabetes if your pancreas can't produce enough insulin to overcome the insulin resistance. Other hormone-related conditions like acromegaly and Cushing syndrome can also cause Type 2 diabetes

Pancreatic Damage: Physical damage to your pancreas — from a condition, surgery or injury — can impact its ability to make insulin, resulting in Type 3c diabetes.

Genetic Mutation: Certain genetic mutations can cause MODY and neonatal diabetes

Long-term use of certain medications can also lead to Type 2 diabetes, including HIV/AIDS medications

COMPLICATIONS OF DIABETES:

Diabetes can lead to acute (sudden and severe) and long-term complications — mainly due to extreme or prolonged high blood sugar levels

Acute Diabetic Complications:

Acute diabetes complications that can be life-threatening include

- **Hyperosmolar hyperglycaemic state[HHS]:** This complication mainly affects people with Type 2 diabetes. It happens when your blood sugar levels are very high (over 600 milligrams per deciliter or mg/dL) for a long period, leading to severe dehydration and confusion. It requires immediate medical treatment
 - **Diabetes related ketoacidosis[DKA]:** This complication mainly affects people with Type 1 diabetes or undiagnosed T1D. It happens when your body doesn't have enough insulin. If your body doesn't have insulin, it can't use glucose for energy, so it breaks down fat instead. This process eventually releases substances called ketones, which turn your blood acidic. This causes labored breathing, vomiting and loss of consciousness. DKA requires immediate medical treatment
 - **Severe low blood sugar [hypoglycaemia]:** Hypoglycemia happens when your blood sugar level drops below the range that's healthy for you. Severe hypoglycemia is very low blood sugar. It mainly affects people with diabetes who use insulin. Signs include blurred or double vision, clumsiness, disorientation and seizures. It requires treatment with emergency glucagon and/or medical intervention
-

Long term diabetes complications:

- Blood glucose levels that remain high for too long can damage your body's tissues and organs. This is mainly due to damage to your blood vessels and nerves, which support your body's tissues
 - Cardiovascular (heart and blood vessel) issues are the most common type of long-term diabetes complication. They include
 - Coronary artery disease
 - Heart attack
 - Stroke
 - Atherosclerosis
 - Other diabetes complications include:
 - Nerve damage (neuropathy), which can cause numbness, tingling and/or pain
 - Nephropathy, which can lead to kidney failure or the need for dialysis or transplant
 - Retinopathy, which can lead to blindness
 - Diabetes related foot conditions
 - Skin infections
 - Gastro paresis
 - Hearing loss
 - Oral health issues like gum disease
 - Depression
 - Amputations
-

DIAGNOSIS:

Healthcare providers diagnose diabetes by checking your glucose level in a blood test. Three tests can measure your blood glucose level

- **Fasting blood glucose test:** For this test, you don't eat or drink anything except water (fast) for at least eight hours before the test. As food can greatly affect blood sugar, this test allows your provider to see your baseline blood sugar
- **Random Glucose test:**Random” means that you can get this test at any time, regardless of if you've fasted.
- **A1c:** This test, also called HbA1C or glycated hemoglobin test, provides your average blood glucose level over the past two to three months
- To screen for and diagnose gestational diabetes, providers order an oral glucose tolerance test

The following test results typically indicate if you don't have diabetes, have prediabetes or have diabetes. These values may vary slightly. In addition, healthcare providers rely on more than one test to diagnose diabetes

Type of test
Fasting blood glucose test
In-range (mg/dL)
Less than 100.
Prediabetes (mg/dL)
100 to 125.
Diabetes (mg/L)
126 or higher.
Random blood glucose test
In-range (mg/dL)
N/A.
Prediabetes (mg/dL)
N/A.
Diabetes (mg/L)
200 or higher (with classic symptoms of hyperglycemia or hyperglycemic crisis).
A1c
In-range (mg/dL)
Less than 5.7%.
Prediabetes (mg/dL)
5.7% to 6.4%.
Diabetes (mg/L)
6.5% or higher.

MANAGEMENT AND TREATMENT:



Diabetes is a complex condition, so its management involves several strategies. In addition, diabetes affects everyone differently, so management plans are highly individualized

The four main aspects of managing diabetes include:

Blood sugar monitoring: Monitoring your blood sugar (glucose) is key to determining how well your current treatment plan is working. It gives you information on how to manage your diabetes on a daily — and sometimes even hourly — basis. You can monitor your levels with frequent checks with a glucose meter and finger stick and/or with a continuous glucose monitor (CGM). You and your healthcare provider will determine the best blood sugar range for you

Oral diabetes medications: Oral diabetes medications (taken by mouth) help manage blood sugar levels in people who have diabetes but still produce some insulin — mainly people with Type 2 diabetes and prediabetes. People with gestational diabetes may also need oral medication. There are several different types. Metformin is the most common.

Insulin: People with Type 1 diabetes need to inject synthetic insulin to live and manage diabetes. Some people with Type 2 diabetes also require insulin. There are several different types of synthetic insulin. They each start to work at different speeds and last in your body for different lengths of time. The four main ways you can take insulin include injectable insulin with a syringe (shot), insulin pens, insulin pumps and rapid-acting inhaled insulin

Diet: Meal planning and choosing a healthy diet for you are key aspects of diabetes management, as food greatly impacts blood sugar. If you take insulin, counting carbs in the food and drinks you consume is a large part of management. The amount of carbs you eat determines how much insulin you need at meals. Healthy eating habits can also help you manage your weight and reduce your heart disease risk.

Due to the increased risk for heart disease, it's also important to maintain a healthy::

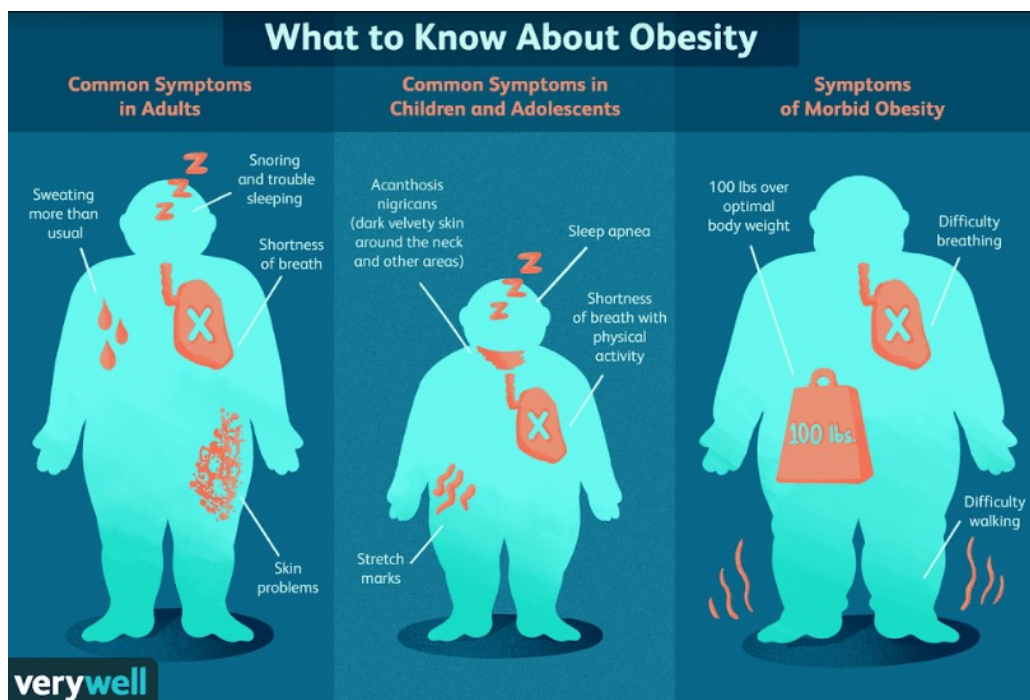
Weight

Blood Pressure

Cholesterol

OBESITY:

Obesity is commonly defined as having too much body mass. A BMI of 30 or higher is the usual benchmark for obesity in adults



ADOLESCENT OBESITY

Although genetics and some disorders cause obesity, most adolescent obesity results from a lack of physical activity and consuming more calories than needed for activity level.

The diagnosis is based on a BMI at or over the 95th percentile for age and gender.

Eating a nutritious diet and increasing exercise help treat obesity, and some children may be given medication for weight loss or weight-loss surgery.

Obesity is defined by the body mass index (BMI). Adolescents whose BMI is in the top 5% for their age and gender are considered to have obesity. Being in the top 5% means their BMI is higher than 95% of their peers (at or over the 95th percentile).

Obesity is twice as common among adolescents as it was 30 years ago. Although most of the complications of obesity occur in adulthood, adolescents with obesity are more likely than their peers to have high blood pressure and type 2 diabetes. Although fewer than one third of adults with obesity were obese as adolescents, most adolescents with obesity become adults with obesity.

Despite many treatment approaches, obesity is one of the most difficult problems to treat, and long-term success rates remain low.

Causes of obesity in Adolescents:

The factors that influence obesity among adolescents are the same as those among adults.

Hormonal disorders, such as an underactive thyroid gland (hypothyroidism) or overactive adrenal glands, may result in obesity but are rarely the cause. Adolescents

with weight gain caused by hormonal disorders are usually short and most often have other signs of the underlying disorder. Any adolescent with obesity who is short and has high blood pressure should be tested for the hormonal disorder Cushing syndrome.

Genetics play a role, which means that some people are at greater risk of obesity than others, and obesity may be more common among members of the same family.

Because of society's stigma against obesity, many adolescents with obesity have a poor self-image and may become socially isolated.

Treatment:

- Healthy eating and exercise habits
- For adolescents 12 years old and older with a BMI at or over the 95th percentile, weight-loss medications
- For adolescents 13 years old and older with a BMI significantly over the 95th percentile, evaluation for weight-loss surgery
- All children and adolescents with obesity are given intensive strategies to help them develop healthy eating and exercise habits and make changes to their health behavior.
- Reducing calorie intake and burning calories are two ways to meet these goals.

Calorie intake is reduced by

- Establishing a well-balanced diet of ordinary foods
- Making permanent changes in eating habits

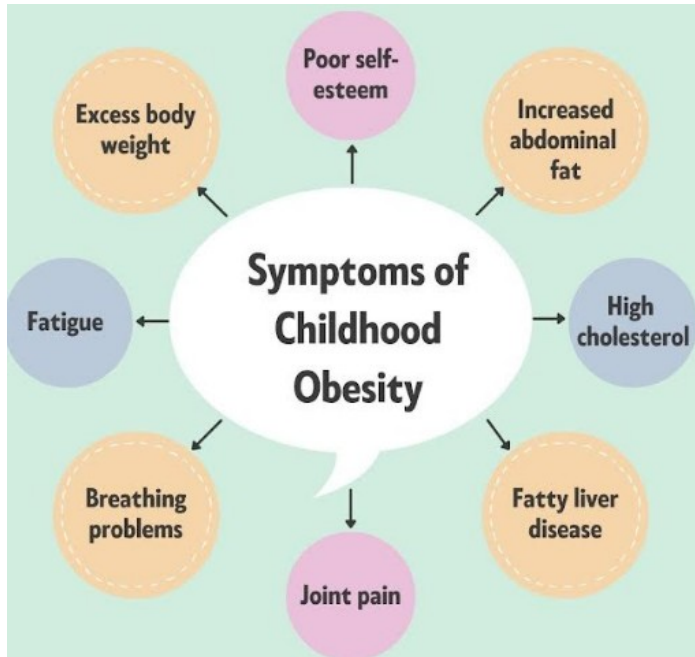
Calorie burning is increased by

- Increasing physical activity
- Adolescents 12 years old and older with obesity (BMI at or over the 95th percentile for age and gender) may be given medications for weight loss.
- Adolescents 13 years old and older with severe obesity (BMI significantly above the 95th percentile for age and gender) may be referred to a surgeon who specializes in this area for a full evaluation for weight-loss surgery (bariatric surgery).
- Counseling to help adolescents cope with social problems, including poor self-esteem, may be helpful.

PAEDIATRIC OR CHILDHOOD OBESITY

Childhood obesity is a serious medical condition that affects children and adolescents. It's particularly troubling because the extra pounds often start children on the path to health problems that were once considered adult problems — diabetes, high blood pressure and high cholesterol. Childhood obesity can also lead to poor self-esteem and depression.

One of the best strategies to reduce childhood obesity is to improve the eating and exercise habits of your entire family. Treating and preventing childhood obesity helps protect your child's health now and in the future



Symptoms

Not all children carrying extra pounds are overweight. Some children have larger than average body frames. And children normally carry different amounts of body fat at the various stages of development. So you might not know by how your child looks if weight is a health concern.

The body mass index (BMI), which provides a guideline of weight in relation to height, is the accepted measure of overweight and obesity. Your child's doctor can use growth charts, the BMI and, if necessary, other tests to help you figure out if your child's weight could pose health problems.

CAUSES

Lifestyle issues — too little activity and too many calories from food and drinks — are the main contributors to childhood obesity. But genetic and hormonal factors might play a role as well.

- **Socioeconomic factors.** People in some communities have limited resources and limited access to supermarkets. As a result, they might buy convenience foods that don't spoil quickly, such as frozen meals, crackers and cookies. Also, people who live in lower income neighborhoods might not have access to a safe place to exercise.
- **Certain medications.** Some prescription drugs can increase the risk of developing obesity. They include prednisone, lithium, amitriptyline, paroxetine (Paxil), gabapentin (Neurontin, Gralise, Horizant) and propranolol (Inderal, Hemangeol).

- overweight. These children are also more likely to develop obstructive sleep apnea, a potentially serious disorder in which a child's breathing repeatedly stops and starts during sleep.
- **Nonalcoholic fatty liver disease (NAFLD).** This disorder, which usually causes no symptoms, causes fatty deposits to build up in the liver. NAFLD can lead to scarring and liver damage.

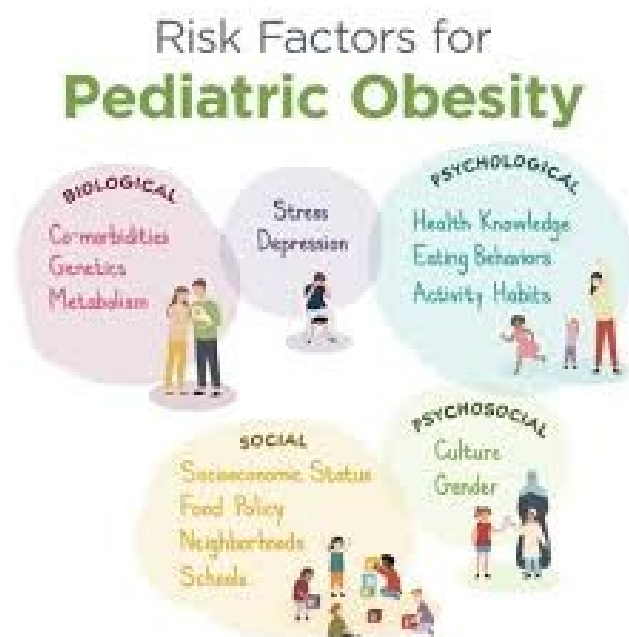
PREVENTION

To help prevent excess weight gain in your child, you can:

- **Set a good example.** Make healthy eating and regular physical activity a family affair. Everyone will benefit and no one will feel singled out.
- **Have healthy snacks available.** Options include air-popped popcorn without butter, fruits with low-fat yogurt, baby carrots with hummus, or whole-grain cereal with low-fat milk.
- **Offer new foods multiple times.** Don't be discouraged if your child doesn't immediately like a new food. It usually takes multiple exposures to a food to gain acceptance.
- **Choose nonfood rewards.** Promising candy for good behavior is a bad idea.
- **Be sure your child gets enough sleep.** Some studies indicate that too little sleep may increase the risk of obesity. Sleep deprivation can cause hormonal imbalances that lead to increased appetite.

Also, be sure your child sees the doctor for well-child checkups at least once a year. During this visit, the doctor measures your child's height and weight and calculates his or her BMI. A significant increase in your child's BMI percentile rank over one year may be a possible sign that your child is at risk of becoming overweight.

Risk factors



WEIGHT CONTROL:

The effort to achieve and maintain a certain weight through diet, exercise, or both is known as weight control. It involves keeping a balance between the energy, or calories, consumed in food and the calories expended in physical activity

Eating healthy foods, staying physically active, and getting quality sleep are examples of healthy lifestyle behaviors that can help you manage your weight and avoid obesity. Managing stress, controlling food portions, and eating mindfully are other behaviors that can help you maintain a healthy weight

What affects weight

Excess weight generally results from an energy imbalance. This is where the energy taken in from eating and drinking is higher than the energy used through physical activity.

But overweight and obesity is a complex issue, influenced by many factors, including:

- genetics
- cultural background
- medical conditions and disability
- mental ill-health – which can affect appetite and motivation to shop and cook healthy food or be active
- some medicines – which can increase appetite or slow the metabolism
- eating disorders
- drugs, tobacco or alcohol
- where people live, go to school and work
- the media
- availability of convenience foods
- portion sizes
- time pressures – which can make it hard to cook healthy meals and exercise
- lifestyle
- shift work – due to disrupted sleep and eating patterns
- inadequate sleep – people who sleep less than 6 hours a night are more likely to gain weight.

Behavior Modification

Changing unhealthy behaviors is crucial for long-term weight management success. Practicing portion control, mindful eating, and keeping a food diary can help you develop a better understanding of your eating patterns. Identifying triggers and adopting healthier alternatives can support sustainable weight loss.

Incorporating Medical Weight Loss

For individuals facing significant challenges in weight management, medically supervised weight loss can provide additional support. You may be wondering, what is medical weight loss? These weight loss programs are designed and supervised by

healthcare professionals specializing in weight management. They offer personalized approaches tailored to individual needs and medical considerations.

Comprehensive Evaluation

How does medical weight loss work? Medical weight loss programs typically start with a comprehensive evaluation. This includes a thorough assessment of medical history, current health conditions, and lifestyle factors. The healthcare provider will analyze factors such as metabolism, body composition, and potential underlying issues that may contribute to weight gain or difficulty losing weight.

Customized Treatment Plans

Based on the evaluation, a customized treatment plan is developed. This plan may include a combination of approaches such as dietary modifications, physical activity recommendations, behavior therapy, and, in some cases, medications or medical procedures. The goal is to address the underlying factors contributing to weight gain and create a sustainable path towards weight management.

Ongoing Support and Monitoring

One of the key advantages of medical weight loss programs is the ongoing support and monitoring provided by healthcare professionals. Regular check-ins, progress tracking, and adjustments to the treatment plan ensure that individuals receive the necessary guidance and support throughout their weight management journey.

BMI

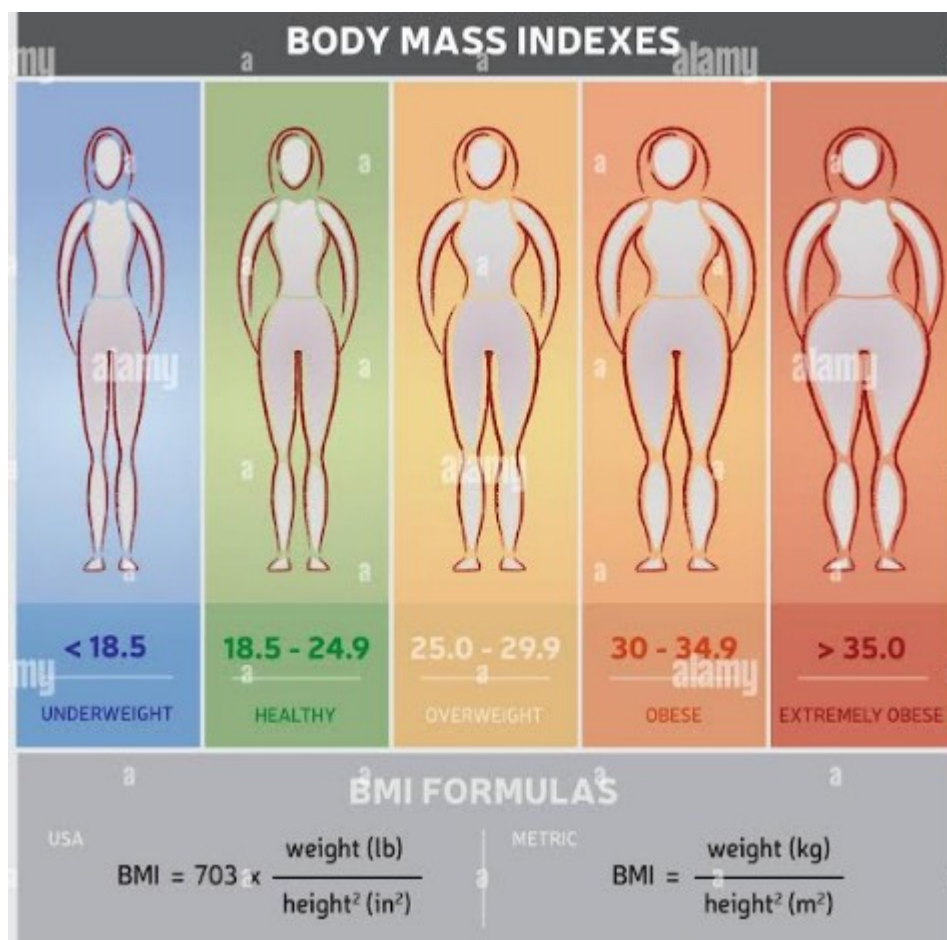
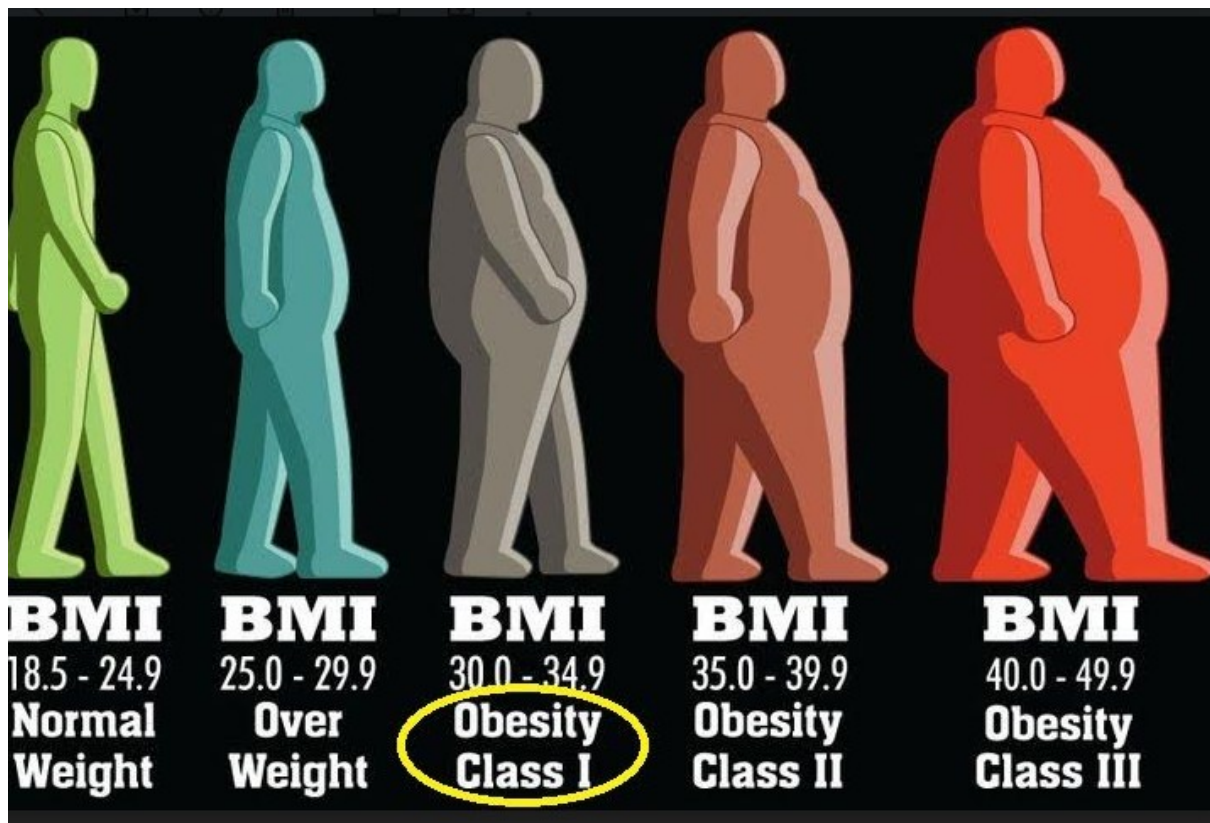
Body mass index. A measure that relates body weight to height. BMI is sometimes used to measure total body fat and whether a person is a healthy weight

Formula: $\text{weight (kg)} / \text{height (m)}^2$

The formula for BMI is weight in kilograms divided by height in meters squared. If height has been measured in centimeters, divide by 100 to convert this to meters.

Example: If a person weighs 65 kg and the person's height is 165 cm (1.65 m), the BMI is calculated as $65 \div (1.65)^2 = 23.87 \text{ kg/m}^2$, which means that the person has a BMI of 23.87 kg/m² and is considered to have a healthy weight.

When calculating in pounds and inches, if a person weighs 120 lbs and the person's height is 5 feet 6 inches (66 inches), the BMI is calculated as $703 \times 120 / (66)^2 = 19.3 \text{ kg/m}^2$, which means that the person has a BMI of 19.3 kg/m² and is considered to have a healthy weight

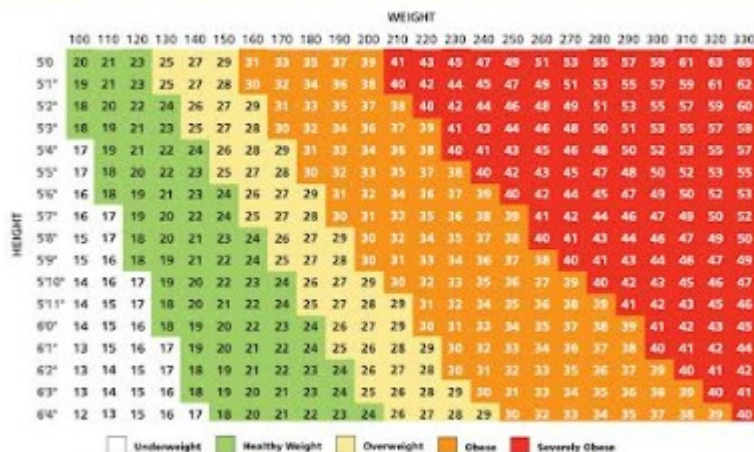


WHO Adult BMI Categories

BMI	Category
< 16.0	Severely Underweight
16.0 - 18.4	Underweight
18.5 - 24.9	Normal
25.0 - 29.9	Overweight
30.0 - 34.9	Moderately Obese
35.0 - 39.9	Severely Obese
≥ 40.0	Morbidly Obese

Are You at a Healthy Weight?

Find the column closest to your weight in pounds. Read down the column until it crosses the row that most closely matches your height in feet and inches. That number is your Body Mass Index. The healthiest BMI range for adults is 18 to 24.



For a healthy weight, ask us how you can eat smart and move more using these simple steps:

- Prepare more meals at home
- Enjoy more fruits and veggies
- Tame the tube: Watch less TV
- Re-think your drink: Choose water
- Choose to move more every day
- Breastfeed your baby
- Right-size your portions



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RESPIRATORY DISEASES

Chronic lung disease, Asthma, COPD; Causes - Breathing pattern (Nasal vs mouth), Smoking – Diagnosis - Pulmonary function testing

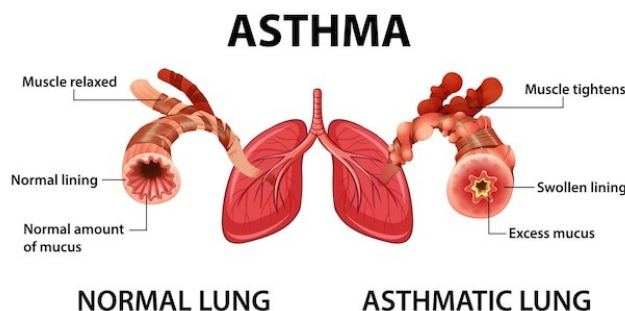
CHRONIC LUNG DISEASE

Chronic lung diseases are a group of conditions that affect the lungs and airways, causing breathing difficulties and other respiratory problems. Some common chronic lung diseases include:

1. Chronic Obstructive Pulmonary Disease (COPD): A progressive lung disease that makes it hard to breathe, caused by long-term exposure to lung irritants like smoke and air pollution.
2. Asthma: A chronic inflammatory disease that causes recurring episodes of wheezing, coughing, chest tightness, and shortness of breath.
3. Pulmonary Fibrosis: A condition where scar tissue builds up in the lungs, making it difficult to breathe and causing fatigue, dry cough, and chest pain.
4. Cystic Fibrosis: A genetic disorder that causes severe damage to the lungs, digestive system, and other organs, leading to respiratory and digestive problems.
5. Emphysema: A type of COPD that damages the air sacs in the lungs, making it hard to breathe and causing shortness of breath.
6. Bronchiectasis: A condition where the airways are damaged, leading to recurring lung infections and chronic coughing.
7. Chronic Bronchitis: A type of COPD that causes inflammation and congestion in the bronchial tubes, leading to coughing and mucus production.

These diseases can be managed with medication, lifestyle changes, and pulmonary rehabilitation, but early diagnosis and treatment are crucial to slow disease progression and improve quality of life.

ASTHMA:



Asthma, also called bronchial asthma, is a disease that affects your lungs. It's a chronic (ongoing) condition, meaning it doesn't go away and needs ongoing medical management.

Asthma is a condition in which your airways narrow and swell and may produce extra mucus. This can make breathing difficult and trigger coughing, a whistling sound (wheezing) when you breathe out and shortness of breath.

For some people, asthma is a minor nuisance. For others, it can be a major problem that interferes with daily activities and may lead to a life-threatening asthma attack

Asthma can't be cured, but its symptoms can be controlled. Because asthma often changes over time, it's important that you work with your doctor to track your signs and symptoms and adjust your treatment as needed

Asthma Attack:

When you breathe normally, muscles around your airways are relaxed, letting air move easily and quietly. During an asthma attack, three things can happen

Bronchospasm: The muscles around the airways constrict (tighten). When they tighten, it makes your airways narrow. Air cannot flow freely through constricted airways

Inflammation: The lining of your airways becomes swollen. Swollen airways don't let as much air in or out of your lungs

Mucus production: During the attack, your body creates more mucus. This thick mucus clogs airways

When your airways get tighter, you make a sound called wheezing when you breathe, a noise your airways make when you breathe out. You might also hear an asthma attack called an exacerbation or a flare-up. It's the term for when your asthma isn't controlled

Types of asthma:

Today, asthma is no longer thought of as a single disease. Asthma is often categorized into different types. They include:

- Allergic asthma
- Aspirin-induced asthma
- Cough-variant asthma
- Exercise-induced asthma
- Nighttime asthma
- Steroid-resistant asthma
- Occupational asthma

Symptoms:

Asthma symptoms vary from person to person. You may have infrequent asthma attacks, have symptoms only at certain times — such as when exercising — or have symptoms all the time.

Asthma signs and symptoms include:

- Shortness of breath

- Chest tightness or pain
- Wheezing when exhaling, which is a common sign of asthma in children
- Trouble sleeping caused by shortness of breath, coughing or wheezing
- Coughing or wheezing attacks that are worsened by a respiratory virus, such as a cold or the flu

Signs that your asthma is probably worsening include:

- Asthma signs and symptoms that are more frequent and bothersome
- Increasing difficulty breathing, as measured with a device used to check how well your lungs are working (peak flow meter)
- The need to use a quick-relief inhaler more often

For some people, asthma signs and symptoms flare up in certain situations:

- **Exercise-induced asthma**, which may be worse when the air is cold and dry
- **Occupational asthma**, triggered by workplace irritants such as chemical fumes, gases or dust
- **Allergy-induced asthma**, triggered by airborne substances, such as pollen, mold spores, cockroach waste, or particles of skin and dried saliva shed by pets (pet dander)

Causes:

- **Family history** :If you have a parent with asthma, you are three to six times more likely to develop asthma than someone who does not have a parent with asthma.
- **Allergies**: Some people are more likely to develop allergies than others, especially if one of their parents has allergies. Certain allergic conditions, such as atopic dermatitis (eczema) or allergic rhinitis (hay fever), are linked to people who get asthma.
- **Viral respiratory infections**: Respiratory problems during infancy and childhood can cause wheezing. Some children who experience viral respiratory infections go on to develop chronic asthma.
- **Occupational exposures** : If you have asthma, exposures to certain elements in the workplace can cause asthma symptoms. And, for some people, exposure to certain dusts (industrial or wood dusts), chemical fumes and vapors, and molds can cause asthma to develop for the very first time.
- **Smoking**: Cigarette smoke irritates the airways. Smokers have a high risk of asthma. Those whose mothers smoked during pregnancy or who were exposed to second hand smoke are also more likely to have asthma.

- **Air Pollution:** Exposure to the main component of smog (ozone) raises the risk for asthma. Those who grew up or live in urban areas have a higher risk for asthma.
- **Obesity:** Children and adults who are overweight or obese are at a greater risk of asthma. Although the reasons are unclear, some experts point to low-grade inflammation in the body that occurs with extra weight. Obese patients often use more medications, suffer worse symptoms and are less able to control their asthma than patients in a healthy weight range.

Testing:

Medical History

You will be asked for some medical history, which should include:

- Family members with asthma
- Allergies
- Smoking and exposure to second hand smoke
- Exposure to pollutants in your workplace

Testing for Asthma

There are also several breathing tests your healthcare provider may perform. The most common lung function test is called spirometry. This lung function test uses a device called, a spirometer, to measure the amount and speed of the air you blow out. This helps your healthcare provider see how well your lungs are working.

Other tests could include allergy testing (blood or skin), a blood test to check for cells responsible for inflammation, exhaled nitric oxide or FeNo test, and challenge tests, such as methacholine. Other lung diseases may cause some of the same symptoms as asthma. If your doctor thinks you might have something else, they may order additional tests.

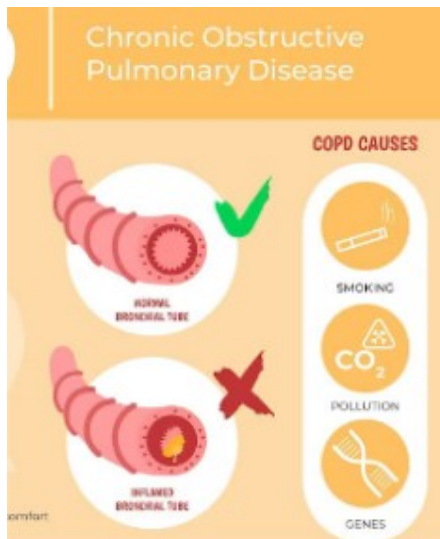
Treatment:

Bronchodilators: These medicines relax the muscles around your airways. The relaxed muscles let the airways move air. They also let mucus move more easily through the airways. These medicines relieve your symptoms

Anti-inflammatory medicines: These medicines reduce swelling and mucus production in your airways. They make it easier for air to enter and exit your lungs. Your healthcare provider may prescribe them to take every day to control or prevent your symptoms of chronic asthma

Biologic therapies for asthma: These are used for severe asthma when symptoms persist despite proper inhaler therapy

COPD : CHRONIC OBSTRUCTIVE PULMONARY DISEASE



Chronic obstructive pulmonary disease (COPD) is a common lung disease causing restricted airflow and breathing problems. It is sometimes called emphysema or chronic bronchitis.

In people with COPD, the lungs can get damaged or clogged with phlegm. Symptoms include cough, sometimes with phlegm, difficulty breathing, wheezing and tiredness. Smoking and air pollution are the most common causes of COPD. People with COPD are at higher risk of other health problems.

Types of COPD

There are two main types of COPD associated with damage to the airways and breathing difficulties. These are called chronic bronchitis and emphysema.

1.Chronic Bronchitis : Chronic bronchitis occurs when the air passages (bronchial tubes) become inflamed and narrowed, restricting airflow through them. As a result, mucus builds up in the airways, which results in chronic cough and mucus (phlegm) production. The combination of narrowed airways and excess mucus makes breathing difficult.

2.Emphysema: Emphysema occurs when air sacs in the lungs are damaged. The air sacs are ordinarily elastic, but damage from exposure to irritants causes them to become saggy and impairs the air sacs' ability to move oxygen and carbon dioxide in and out of the lungs.

Most people with COPD have both chronic bronchitis and emphysema.

Risk Factors:

Certain risk factors can increase a person's risk of developing COPD:

- Smoking or exposure to second-hand smoke
- Occupational exposure to chemicals, dust, vapors, or fumes
- Indoor and outdoor air pollutants such as smog, mold, dust mites, asbestos, and combustible pollutants (e.g., firewood smoke, furnaces that use gas or coal)
- Frequent or severe respiratory infections in childhood that stunt lung growth
- Asthma in childhood

- Alpha-1 antitrypsin deficiency, a rare genetic condition

Signs and symptoms of COPD :

COPD is not curable but symptoms can improve if one avoids smoking and exposure to air pollution and gets vaccines to prevent infections. It can also be treated with medicines, oxygen and pulmonary rehabilitation.

Signs and symptoms of COPD may include:

- Shortness of breath, especially during physical activities
- Wheezing
- Chest tightness
- A chronic cough that may produce mucus (sputum) that may be clear, white, yellow or greenish
- Frequent respiratory infections
- Lack of energy
- Unintended weight loss (in later stages)
- Swelling in ankles, feet or legs

People with COPD are also likely to experience episodes called exacerbations, during which their symptoms become worse than the usual day-to-day variation and persist for at least several days.

Causes

Several processes can cause the airways to become narrow and lead to COPD. There may be destruction of parts of the lung, mucus blocking the airways, and inflammation and swelling of the airway lining.

COPD develops gradually over time, often resulting from a combination of risk factors:

- tobacco exposure from active smoking or passive exposure to second-hand smoke;
- occupational exposure to dusts, fumes or chemicals;
- indoor air pollution: biomass fuel (wood, animal dung, crop residue) or coal is frequently used for cooking and heating in low- and middle-income countries with high levels of smoke exposure;
- early life events such as poor growth in utero, prematurity, and frequent or severe respiratory infections in childhood that prevent maximum lung growth;
- asthma in childhood
- a rare genetic condition called alpha-1 antitrypsin deficiency, which can cause COPD at a young age.

COPD should be suspected if a person has typical symptoms, and the diagnosis confirmed by a breathing test called spirometry, which measures how the lungs are working. In low- and middle-income countries, spirometry is often not available and so the diagnosis may be missed.

Diagnosis:

To assess your lungs and overall health, your healthcare provider will take your medical history, perform a physical exam and order some tests, like breathing tests

Medical History:

To diagnose COPD, your provider will ask questions like

- Do you smoke?
- Have you had long-term exposure to dust or air pollutants?
- Do other members of your family have COPD?
- Do you get short of breath with exercise? When resting?
- Have you been coughing or wheezing for a long time?
- Do you cough up phlegm?

Physical Exam:

To help with the diagnosis, your provider will do a physical exam that includes:

- Listening to your lungs and heart
- Checking your blood pressure and pulse
- Examining your nose and throat
- Checking your feet and ankles for swelling

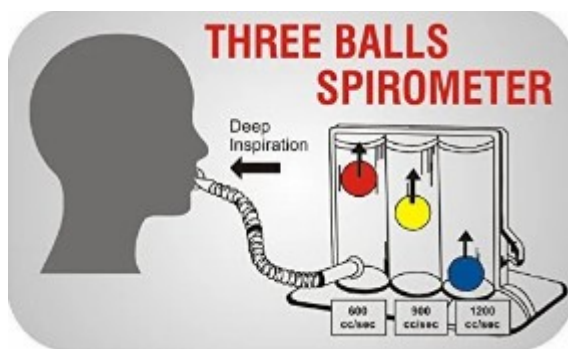
Tests:

Providers use a simple test called spirometry to see how well your lungs work. For this test, you blow air into a tube attached to a machine. This lung function test measures how much air you can breathe out and how fast you can do it

If you have symptoms of COPD, your healthcare provider will investigate to determine the cause. They may ask for a detailed personal and family medical history and perform a physical examination. During the exam, they will use a stethoscope to listen to your lungs as you breathe to check for crackling or wheezing sounds. 10

In addition to the physical exam, your provider may order additional procedures and tests to diagnose COPD, including:

- **Spirometry:** A non-invasive lung function test that measures the maximum amount of air you can hold in your lungs and how quickly you can blow the air out.



- **Six-minute walk test:** You will walk for 6 minutes at your usual pace while doctors measure your blood pressure, pulse, and oxygen levels to check on how well your lungs function.
- **Complete blood count (CBC):** A blood test that helps your doctor determine if you have an infection or low iron (anemia).

- **Arterial blood gas (ABG):** A blood test that measures how much oxygen and carbon dioxide are in your blood to determine how well your lungs are functioning.
- **Alpha-1 antitrypsin (AAT) deficiency:** A genetic test that checks for levels of AAT in the blood. Low levels of AAT may be a sign of a rare genetic condition called AAT deficiency, which is linked to an increased risk of COPD.
- **Computed tomography (CT) scan:** An imaging test that takes detailed pictures of your lungs and other structures in your chest so doctors can look for damage and scarring in the airways.
- **Chest X-ray:** An imaging test that takes pictures of the structures in and around your chest to check for signs of lung tissue damage and scarring.

Stages:

Mild COPD (stage 1 or early stage)

The first sign of COPD is often feeling out of breath with light exercises, like walking up stairs. Because it's easy to blame this symptom on being out of shape or getting older, many people don't realize they have COPD. Another sign is a phlegmy cough (a cough with mucus) that's often particularly troublesome in the morning. These are early warning signs of COPD

Moderate to severe COPD (stages 2 and 3)

In general, shortness of breath is more evident with more advanced COPD. You may develop shortness of breath even during everyday activities. Also, exacerbations of COPD — times when you experience increased phlegm, discoloration of phlegm, and more shortness of breath — are generally more common in higher stages of COPD. You also become prone to lung infections like bronchitis and pneumonia.

Very severe COPD (stage 4)

When COPD becomes severe, almost everything you do can cause shortness of breath. This limits your mobility. You may need supplemental oxygen from a portable tank

Treatments for COPD

The goals of treating COPD include improving airflow to make breathing easier, preventing disease progression, reducing exacerbations and complications, and improving quality of life.

Treatments vary depending on the severity of the condition but generally include a combination of lifestyle interventions, prescription medications, oxygen therapy, and pulmonary rehabilitation. Surgery may be an option for some people with severe COPD.

1. Lifestyle Interventions:

Making healthy lifestyle choices is an essential part of treating COPD. Lifestyle interventions for COPD include:

- Quitting smoking
- Limiting exposure to lung irritants: secondhand smoke, pollutants, gasses, and fumes
- Regular exercise
- Eating a healthy, balanced diet
- Getting vaccinated for influenza, pneumococcal viruses, and COVID-19

- Avoiding contact with people who have respiratory infections

2. Prescription Medications:

Medications to reduce inflammation and swelling to open the airways, decrease mucus production, and improve lung function are often prescribed to treat COPD. Many drugs are breathed in through inhalers or nebulizer devices, and some are taken orally as pills. Prescription medicines for COPD include:

Bronchodilators: These relax the muscles around the airways to make breathing easier. Some take a while to work but last about 12 to 24 hours, and others are fast-acting to help relieve sudden shortness of breath.

Corticosteroids: These reduce inflammation and swelling to open the airways. Corticosteroids are usually inhaled, but pills may be prescribed during a flare-up.

Antibiotics: These medications treat bacterial respiratory infections to help clear the infection and prevent complications.

3. Oxygen therapy:

Supplemental oxygen may be recommended if you have low blood oxygen levels when at rest, during physical activities, or while sleeping.

4. Pulmonary Rehabilitation :

Pulmonary rehabilitation is a supervised program that includes physical exercise, education, and support. These programs can help you improve your endurance and breathing ability during exercise and daily activities. Pulmonary rehab programs are generally offered in group settings, so you can meet others with COPD to give and receive support.

5. Surgery

Surgery may be an option for some people with severe and frequent symptoms that are not well-managed with medications. Surgical procedures improve lung capacity and make it easier to breathe. Lung surgeries for COPD include:

- **Bullectomy:** Bullae are large air pockets that form inside the lung due to certain medical conditions, such as emphysema and chronic obstructive pulmonary disease. Bullae often grow in size once they form. Without treatment, enlarged bullae take up an increasing amount of space in the lungs, causing pressure to build. The pressure can make it difficult for a person to breathe. Bullectomy, which is the surgical removal of the bullae, if they cause breathing problems or other health complications
- **Lung volume reduction surgery :** Surgeon removes areas of diseased lung tissue that are not being used by the body. Then the remaining lung tissue can work better. After surgery, people often have less shortness of breath and better quality of life. They also are better able to exercise.
- **Lung transplant:** **Lung transplantation, or pulmonary transplantation,** is a surgical procedure in which one or both lungs are replaced by lungs from a donor. Donor lungs can be retrieved from a living or deceased donor. A living donor can only donate one lung lobe.

DIFFERENCE BETWEEN ASTHMA AND COPD

	COPD	ASTHMA
Smoker or ex-smoker	Most	Possibly
Symptoms under age 35	Rare	Common
Atopy	Uncommon	Common
Cellular infiltrate	Macrophages Neutrophils CD8+ T cells	Eosinophils CD4+ T cells
Cough and sputum	Daily/common	Intermittent
Breathlessness	Persistent/progressive	Variable
Night time symptoms	Uncommon	Common
Significant diurnal variability of symptoms	Uncommon	Common
Bronchodilator response (FEV1, PEFr)	<15%	>20%
Corticosteroid response	Poor	Good

NOSE AND MOUTH BREATHING

Mouth Breathing	Nose Breathing
Agitated mind	Calm mind
Poor sleep	Deep sleep/get to sleep more easily
Daytime fatigue	Wake up feeling fresh
Brain fog	Clarity of mind
Lack of focus	Laser focus
Poor concentration	Good concentration
= <u>Every day is a struggle</u>	= <u>Every day is inspiring</u>

Benefits Of Nose Breathing Vs Mouth Breathing

1. The inside of your nose is lined with tiny hairs called cilia. The cilia filter, warm and humidify inhaled air. They protect you against 20 billion particles of foreign matter every day.
2. Some people argue you need to mouth breathe to bring in enough air during exercise. But research proves the opposite. The extra resistance during nose breathing results in 10 to 20% better oxygenation
3. Nose breathing is 22% more efficient than mouth breathing . Which means breathing is much easier, and the breathing muscles don't get so tired.

4. Nose breathing keeps your airways clear. Breathing in through the nose and out through the mouth causes nasal congestion and makes it harder to breathe . If you're wondering how to breathe with a stuffy nose, the surprising answer is that you need to get used to nose breathing.
 5. The lungs draw oxygen from inhaled air during nasal exhalation too. Nose breathing is slower, and nasal exhalation creates a backflow of air into the lungs. The air stays in the lungs for longer, and the body has more time to extract oxygen from that air.
 6. The air hunger during nose breathing forces you to slow down until your breath is properly trained. This reduces stress and high blood pressure. And it stops you from overtraining.
 7. Nose breathing harnesses nitric oxide (NO) Nitric oxide is produced in the sinuses around the nose. It protects against airborne viruses, bacteria, allergens, and other pathogens. It opens the blood vessels in the lungs, allowing better oxygen diffusion.
 8. When there is a proper balance of oxygen and carbon dioxide (CO₂) in the blood, the body maintains a balanced pH. Most oxygen travels around the body in red blood cells. CO₂ is the catalyst that causes the red blood cells to release this load of oxygen to the body. If you breathe through an open mouth, you lose too much CO₂ . So nose breathing enhances your body's ability to access the oxygen in your blood.
 9. Nose breathing during sleep makes sleep apnea less severe. Scientists found that during mouth breathing, apneas increase, and oxygen desaturation is worse.
 10. It is not possible to maintain diaphragm breathing through the mouth. Nose breathing engages and strengthens your diaphragm, which is an integral muscle of the core and necessary for the stability of your spine.
 11. During exercise, nose breathing can improve mental focus and boost coronary artery blood flow
 12. Nose breathing reduces symptoms of exercise-induced asthma because it protects the airways from trauma and inflammation.
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SMOKING

Smoking cigarettes can kill you, but before you die, you could experience some pretty terrible diseases and health conditions from smoking. Here are some of the most gruesome diseases caused by smoking*:

1. Lung Cancer
 2. COPD (chronic obstructive pulmonary disease)
 3. Heart Disease
 4. Stroke
 5. Asthma
 6. Reproductive Health in Women
 7. Premature, Low Birth-Weight Babies
 8. Diabetes
 9. Blindness, Cataracts and Age-Related Macular Degeneration
 10. Over 10 Other Types of Cancer, Including Colon, Cervix, Liver, Stomach and Pancreatic Cancer
-

PULMONARY FUNCTION TESTS (PFTS)

Pulmonary function tests (PFTs) are tests that show how well your lungs are working. The tests measure lung volume, capacity, rates of flow, and gas exchange. This information can help your healthcare provider diagnose and decide the treatment of certain lung disorders.

There are several types of disorders that cause problems with air moving in and out of the lungs:

- **Obstructive.** This is when air has trouble flowing out of the lungs because of airway resistance. This causes slower flow of air.
- **Restrictive.** This is when the lung tissue or chest muscles can't expand enough. This creates problems with air flow, mostly because you have less lung volume.

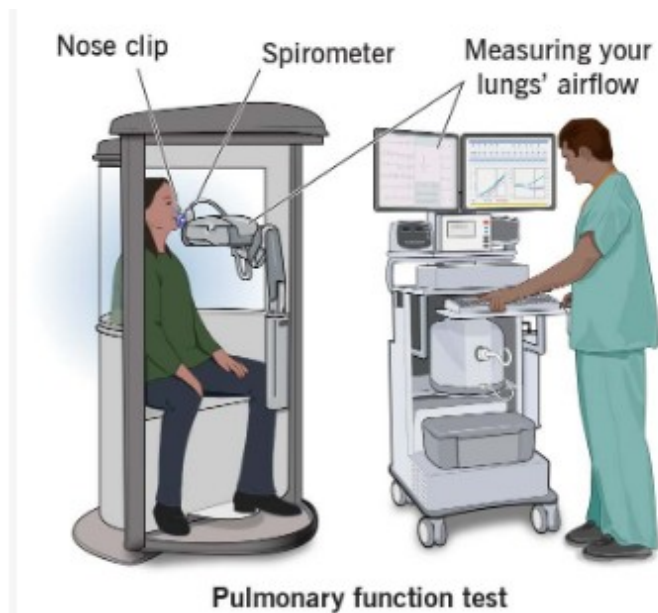
Types of PFT:

PFTs can be done in two ways. These two ways may be used together and do different tests. It depends on the information that your healthcare provider is looking for:

Spirometry. A spirometer is a device with a mouthpiece hooked up to a small electronic machine.



Plethysmography. You sit or stand inside an airtight box to do the tests.



PFTs measure:

- Tidal volume. This is the amount of air breathed in or out during normal breathing.
- Minute volume. This is the total amount of air breathed out per minute.
- Vital capacity. This is the total volume of air that can be breathed out after breathing in as much as you can.
- Functional residual capacity. This is the amount of air left in lungs after breathing out normally.
- Residual volume. This is the amount of air left in the lungs after breathing out as much as you can.
- Total lung capacity. This is the total volume of the lungs when filled with as much air as possible.
- Forced vital capacity (FVC). This is the amount of air breathed out forcefully and quickly after breathing in as much as you can.
- Forced expiratory volume. This is the amount of air breathed out during the first, second, and third seconds of the FVC test.
- Forced expiratory flow. This is the average rate of flow during the middle half of the FVC test.
- Peak expiratory flow rate. This is the fastest rate that you can force air out of your lungs.

Normal values for PFTs vary from person to person. The amount of air breathed in and out in your test results are compared with the average for someone of the same age, height, sex, and race. Results are also compared with any of your past test results. You may need other tests if you have abnormal PFT measurements or if your results have changed.

Why might I need pulmonary function tests?

There are many different reasons why pulmonary function tests (PFTs) may be done. They are sometimes done in healthy people as part of a routine physical. They are also

routinely done in certain types of work environments to make sure of employee health. This might be in graphite factories or coal mines. Or you may have PFTs if your healthcare provider needs help to diagnose you with a health problem. These include:

- Allergies
- Respiratory infections
- Trouble breathing from injury to the chest or a recent surgery
- Long-term (chronic) lung conditions, such as asthma, bronchiectasis, emphysema, or chronic bronchitis
- Asbestosis, a lung disease caused by inhaling asbestos fibers
- Restrictive airway problems from scoliosis, tumors, or inflammation or scarring of the lungs
- Sarcoidosis, a disease that causes lumps of inflammatory cells around organs, such as the liver, lungs, and spleen
- Scleroderma, a disease that causes thickening and hardening of connective tissue

PFTs may be used to check lung function before surgery or other procedures. This may be done in people who have lung or heart problems, who are smokers, or who have other health conditions. Another use of PFTs is to assess treatment for asthma, emphysema, and other chronic lung problems. Your healthcare provider may also have other reasons to advise PFTs.

Risks of pulmonary function tests

Because pulmonary function testing is not an invasive procedure, it is safe and quick for most people. But the person must be able to follow clear, simple directions. All procedures have some risks.

The risks of this procedure may include:

- Dizziness during the tests
- Feeling short of breath
- Coughing
- Asthma attack brought on by deep inhalation

In some cases, you shouldn't have PFTs. Reasons for this can include:

- Recent eye surgery, because of increased pressure inside the eyes during the procedure
- Recent belly or chest surgery
- Chest pain, recent heart attack, or an unstable heart condition
- A bulging blood vessel (aneurysm) in the chest, belly, or brain
- Active tuberculosis or respiratory infection, such as a cold or the flu

Certain things can make PFTs less accurate. These include:

- Not following the instructions for doing the test
- Using medicines that open the airways (bronchodilators)
- Using pain medicines
- Pregnancy

- Stomach bloating. This can affect your ability to take deep breaths
 - Extreme tiredness or other conditions, such as a head cold, that affect your ability to do the tests
-